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# **The role of requirements engineering in defining the explainability of AI systems**

From AI ethical guidelines to requirements

Nagadivya Balasubramaniam

A doctoral thesis completed for the degree of Doctor of Science (Technology) to be defended, with the permission of the Aalto University School of Science, at a public examination held at the lecture hall [name of hall] of the school on dd Month yyyy at [time of day].

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The use of artificial intelligence (AI) is expanding rapidly, but the growing adoption of AI systems has raised ethical concerns among organizations and other stakeholders. Many AI systems are difficult to understand because of their black-box nature. This has led researchers and regulators, most notable through the EU AI Act, to emphasize transparency and explainability as important requirements for ethical and responsible AI development. However, there is still limited understanding of how explainability requirements are defined and implemented in practice, and how requirements engineering (RE) can support this process.

This thesis investigates how RE can support the definition of explainability of AI systems. The results are based on expert interviews, document analysis, workshops, and a literature review. We analyzed the AI ethical guidelines of sixteen organizations, interviewed practitioners and industry experts, reviewed existing literature, and conducted workshops with practitioners.

One of our main findings is that organizations consistently emphasize accountability, explainability, fairness, privacy, and transparency in their AI ethical guidelines and explainability is closely tied to transparency and trustworthiness. Industry experts view explainability as essential for helping users understand AI systems, their risks, and their limitations, and for supporting accountability and human oversight. However, customers often remain less aware about the importance of explainability than practitioners. Furthermore, organizations face practical challenges, such as, limited resources and complexity of the models used, when incorporating explainability into AI system development.

Based on these findings, this thesis proposes a model of explainability components to support the systematic identification of explainability needs, along with a structured template for representing the explainability requirements of AI systems. In addition, the thesis proposes a set of good practices, such as clearly defining the purpose of an AI system, having multidisciplinary teams, and analyzing potential risks and negative consequences of AI systems. Building on these contributions, this thesis recommends a lightweight approach consisting of principles, practices, and guiding questions to help organizations address the transparency requirements of the EU AI Act using explanations.

The main conclusion of this work is that explainability is not only an ethical and regulatory concern, but also a quality requirement that can be systematically defined using established RE practices. The main implications of this thesis are twofold. For researchers, this thesis provides empirically grounded models and practices that address the gap between theoretical (candidate) solutions and their real-world evaluation. For practitioners, this thesis offers concrete tools for defining explainability requirements and practical guidance for addressing the transparency requirements of the EU AI Act.

# Acknowledgements

I want to thank many people

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# List of Publications

This doctoral thesis consists of the following publications which are referred in the text by their respective Roman numerals:

## **I - Ethical guidelines for solving ethical issues and developing AI systems**

N. Balasubramaniam, M. Kauppinen, S. Kujala, and K. Hiekkanen, “Ethical Guidelines for Solving Ethical Issues and Developing AI Systems,” in *Product-Focused Software Process Improvement*, vol. 12562, pp. 331–346, 2020. doi: [10.1007/978-3-030-64148-1\\_21](https://doi.org/10.1007/978-3-030-64148-1_21).

## **II – Transparency and explainability of AI systems: From ethical guidelines to requirements**

N. Balasubramaniam, M. Kauppinen, A. Rannisto, K. Hiekkanen, and S. Kujala, “Transparency and explainability of AI systems: From ethical guidelines to requirements,” *Information and Software Technology*, vol. 159, p. 107197, 2023, doi: [10.1016/j.infsof.2023.107197](https://doi.org/10.1016/j.infsof.2023.107197).

## **III – Candidate Solutions for Defining Explainability Requirements of AI Systems**

N. Balasubramaniam, M. Kauppinen, H.-L. Truong, and S. Kujala, “Candidate Solutions for Defining Explainability Requirements of AI Systems,” in *Requirements Engineering: Foundation for Software Quality*, vol. 14588, 2024, pp. 129–146. doi: [10.1007/978-3-031-57327-9\\_8](https://doi.org/10.1007/978-3-031-57327-9_8).

## **IV – Explainability of AI system in practice: reasons, challenges, and role in the EU AI Act**

N. Balasubramaniam, M. Kauppinen, T. Männistö, H.-L. Truong, and S. Kujala, *\*submitted in a Journal, 2026*

# Author's Contribution

## **I - Ethical guidelines for solving ethical issues and developing AI systems**

Balasubramaniam was responsible for conducting the study, including the problem definition, study design, data collection and analysis, and reporting. The second author participated actively in the problem definition, data collection and analysis and provided continuous feedback on the report. The third and fourth author had a role of supporting and commenting throughout the study.

## **II – Transparency and explainability of AI systems: From ethical guidelines to requirements**

Balasubramaniam was responsible for conducting the study, including the problem definition, the study design, data collection and analysis, and reporting. The second author participated actively in the problem definition, study design, data collection and analysis and provided continuous feedback on the report. The third author assisted the data collection, data analysis, and commented on parts of the work of phase 2 study. The fourth and fifth authors participated in study design and provided overall comments for the paper.

## **III – Candidate Solutions for Defining Explainability Requirements of AI Systems**

Balasubramaniam had a primary role in study designing, analyzing, and reporting. The second author actively reviewed the study design, data analysis, and reporting. The third author participated in study design and commented on other parts of the work. The fourth author commented on final article and provided suggestion for introduction and research method sections.

## **IV – Explainability of AI system in practice: reasons, challenges, and role in the EU AI Act**

Balasubramaniam had the main responsibility in study design, data collection and analysis, and reporting. The second author participated in the data collection and supported the data analysis. The second and third authors were involved during the creation of lightweight approach to address the EU AI Act's transparency requirements and provided continuous



feedback for different sections of the paper. The fourth and fifth authors had a role of reviewing and commenting throughout the study.

The CRediT (Contribution Roles Taxonomy) model (Brand et al., 2015) was used to clarify my contributions to the included papers of this thesis.

Table: Contributions of Nagadivya Balasubramaniam to the appended papers of this thesis

| <b>Role/Paper</b>          | <b>I</b> | <b>II</b> | <b>III</b> | <b>IV</b> |
|----------------------------|----------|-----------|------------|-----------|
| Conceptualization          | ✓        | ✓         | ✓          | ✓         |
| Methodology                | ✓        | ✓         | ✓          | ✓         |
| Investigation              | ✓        | ✓         | ✓          | ✓         |
| Validation                 | ✓        | ✓         | ✓          | ✓         |
| Data curation              | ✓        | ✓         | ✓          | ✓         |
| Writing – original draft   | ✓        | ✓         | ✓          | ✓         |
| Writing – review & editing | ✓        | ✓         | ✓          | ✓         |
| Visualization              | ✓        | ✓         | ✓          | ✓         |
| Project administration     | ✓        | ✓         |            | ✓         |
| Supervision                |          |           |            |           |
| Funding acquisition        |          |           |            |           |

# 1 Introduction

As the development of AI technologies brought its fair share of ethical concerns, organizations are committing to develop ethical and responsible artificial intelligence (AI) systems. The need is more especially for high-risk AI system and AI systems used in critical domains such as banking and autonomous transportation. Developing ethical and responsible AI systems does not just account developing trustworthy, transparent, understandable, and explainable AI models rather making the system decisions, purpose, and risks understandable for the end-users as well.

Requirements engineering (RE) is a branch of software engineering that contains models, methods, and practices that enables organizations in defining and documenting requirements of the system. This thesis investigates how organizations can use requirements engineering in developing explainable AI systems.

## 1.1 Motivation and Background

AI technologies have delivered significant social benefits (Rahwan, 2018). However, their black-box nature raises ethical concerns among stakeholders around safety, privacy, security, and transparency (Peslak et al., 2004, Vampley et al. 2018). Ethical issues like bias, diversity, privacy preferences need to be addressed throughout the software engineering (SE) process (Jones et al., 2015). Aydemir and Dalpiaz (2018) shares that ethical issues should be considered from the very start of the SE process, beginning with requirements definition.

Algorithmic decision-making is now common in critical areas, such as loan processing, criminal identification, and cancer detection (Abdollahi & Nasraoui, 2018, Habibullah & Horkoff, 2021). More organizations are building AI systems whereas the black-box nature of systems creates ethical concerns. To address these concerns, groups like IEEE and ACM have created ethical guidelines for responsible AI. Frost et al. (2022) and Jobin et al. (2019) reviewed 47 and 84 AI ethical guidelines respectively and both reviews found that transparency is the most common ethical guideline, and explainability is often seen as part of transparency. Many organizations also create their own AI ethical guidelines based on issues that are relevant to them.

Organizations use different machine learning (ML) models and algorithms for decision-making. However, the outputs of these models are often hard to understand and lack transparency (Chazette and Schneider, 2020). Key studies in requirements engineering (RE) research area (Chazette et al. 2021, Chazette and Schneider, 2020) identify explainability as a key requirement that improves transparency. Ahmad et al.'s (2020) study on requirements engineering techniques found that explanations build users trust and improve decision-making.

Transparency and explainability are seen as key quality requirements of AI systems (Chazette et al. 2021a, Chazette and Schneider, 2020, Cysneiros et al., 2018). According to Chazette et al. (2021), explainability can affect user needs, cultural values, laws, corporate values, and other quality aspects of AI systems. The number of studies focusing on transparency and explainability is increasing in research areas such as RE, SE, explainable AI (XAI), and HCAI (human-centered AI) but the research on how to incorporate explainability and how to define the explainability requirements of AI systems require are preliminary and need more investigation.

In recent times, more organizations are using AI to automate tasks and support decision-making process (Bruijn et al., 2022). Currently, Generative AI (Gen AI) offers promising benefits, such as higher productivity, cost and time savings in software development (Sauvola et al. 2024). However, LaGrandeur (2024) warns about the exaggeration of AI capabilities, which could harm the society. To address this uncertainty, the European Union proposed the AI Act (AI Act) to promote human-centric and trustworthy AI (AI Act, 2024). As AI adoption grows it is important to examine both current and future risks (Benjio et al., 2024), and these risks include discrimination (Benjio et al., 2024, Arora et al., 2023, Slattery et al., 2026), privacy and security threats (Benjio et al., 2024, Cheatham et al., 2019, Slattery et al., 2026), misinterpretation (Arora et al., 2023), existential threats (Bucknall & Dori-Hacohen, 2022, Hendrycks et al., 2023) and environmental harms (Slattery et al., 2026).

In software engineering, studies on ethical AI development identify explainability as an essential quality requirement (Vogelsang & Borg, 2019, Chazette et al. 2022). Defining explainability requirements is seen as important for building transparency and trustworthy AI systems (Habiba et al. 2025). Explainability can also affect other quality requirements, such as trustworthiness, privacy, and fairness (Habiba et al. 2025, Chazette et al. 2022). Chazette et al. (2022) found that good explanations improve user satisfaction and system acceptance.

Research on explainability is growing across many fields (Lewis et al. 2021). However, in requirements engineering, there is a limited understanding of how explainability works in practice (Vogelsang & Borg, 2019, Habiba et al. 2025, Habibullah et al. 2023, Droste et al. 2024, Chazette et al. 2022). Panigutti et al.

(2023) studies XAI in relation to the AI Act and linked it to the Act's transparency and humans oversight requirements. However, two studies (Habiba et al. 2025, Panigutti et al. 2023) emphasized that there is a need for more research to understand how explainability is addressed in the AI Act and how it can improve transparency in practice.

The aim of this thesis is to investigate how requirements engineering can support defining explainability of AI systems. One of the key focuses is on identifying how to define explainability requirements of AI system in real projects. In addition, we also investigate what is the current state of explainability in practice. Moreover, the interest also lies in understanding how explainability can support organizations in adopting the AI Act in AI systems they develop, especially the transparency requirements of the AI Act.

## 1.2 Research Problem and Questions

In this thesis, we aim to explore the following research problem:

***How requirements engineering support organizations in defining the explainability (requirements) of AI systems?***

The particular interest of the study is in explainability of AI systems. To our knowledge, the RE literature holds very less empirical study on explainability and explainability requirements of AI systems.

The research is based on the increased interest in organizations to develop ethical and trustworthy AI systems and to satisfy regulatory obligations. Transparency is a key aspect often associated with developing ethical AI systems and explainability is seen as a core part of ensuring transparency of AI systems. As the topic is quite new and fresh the opportunity to explore the topic from the perspective of explainability seemed timely and essential. Recently, the adoption of the AI Act in European Union mandated all high-risk systems developed to address requirements such as transparency and human oversight. Therefore, the aim is to first study the current state of explainability in practice and explore how requirements engineering can help organizations in developing explainable AI systems which in turn can support transparency requirements of the AI Act.

To investigate our research problem, we formulated the following research questions:

- **RQ1:** *What quality requirements do organizations emphasize in AI ethical guidelines?*
- **RQ2:** *How do industry experts perceive the explainability of AI systems in practice?*

- **RQ3:** How can explainability of AI systems address the transparency requirements of the EU AI Act?
- **RQ4:** What good RE practices can support defining the explainability of AI systems?

The following table shows the relationship between the research questions and the publications of the thesis.

Table 1. Relation of the research questions and the publications

| ID | Research question                                                                            | Publications |    |     |    |
|----|----------------------------------------------------------------------------------------------|--------------|----|-----|----|
|    |                                                                                              | I            | II | III | IV |
| 1  | What quality requirements do organizations emphasize in AI ethical guidelines?               | x            | x  |     |    |
| 2  | How do industry experts perceive the explainability of AI systems in practice?               |              | x  |     | x  |
| 3  | How can explainability of AI systems address the transparency requirements of the EU AI Act? |              |    |     | x  |
| 4  | What good RE practices can support defining the explainability of AI systems?                | x            | x  | x   | x  |

Table 2. Research objectives of the publications

| ID  | Name of the publication                                                                  | Objective                                                                                                                                                                     |
|-----|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I   | Ethical guidelines for solving ethical issues and developing AI systems                  | Analyze ethical guidelines developed by organizations to solve potential ethical issues of AI systems                                                                         |
| II  | Transparency and explainability of AI systems: from ethical guidelines to requirements   | Explore the ethical guidelines in organizations to identify key quality requirements of AI systems and investigate how explainability requirements can be defined in practice |
| III | Candidate solutions for defining explainability requirements of AI systems               | Investigate the solutions proposed in the literature for defining the explainability requirements of AI systems                                                               |
| IV  | Explainability of AI systems in practice: reasons, challenges, and role of the EU AI Act | Understand how industry experts perceive explainability and demonstrate how explainability can address the transparency requirements of the AI Act                            |

### **1.3 Research scope and focus**

The research demonstrated in this thesis was carried out within the disciplines of software engineering, which covers the entire software development lifecycle. The research focuses on requirements engineering, which comes under the umbrella of software engineering.

The particular interest of the thesis lies in exploring how requirements engineering can enable defining explainability of AI systems. The research started by analyzing the AI ethical guidelines as foundation as they are the first step that organizations defined to develop ethical AI systems.

Explainable AI (XAI) is a research discipline that is actively researching on developing explainable models and algorithms. This study does not focus only on technical perspective of developing explainability of AI system but covers explainability throughout the AI system lifecycle. Similarly, ethics field classifies transparency and explainability as key requirements for developing ethical AI systems. As mentioned earlier, we start our study basing on the AI ethical guidelines and explore how transparency and explainability requirements can be defined for AI systems.

All in all, this thesis primarily focuses on requirements engineering. The particular interest is defining explainability requirements and utilizing requirements engineering practices when developing explainability of AI systems. The research identifies existing RE practices that can support explainability definition and identify new RE practices from the experiences of industry experts.

### **1.4 Structure of the thesis**

This thesis consists of a summary part and the research articles. Section 2 provides an overview of literature that gives background and are relevant to this thesis. In Section 3, the research goals, research questions, and methodologies used for the empirical studies and literature review are summarized. The results of this work, structured by the research goals, are presented in Section 4. We discuss the results, threats to validity and limitations of this thesis in Section 5. Finally, Section 6 concludes this thesis work. The four publications annexed to the thesis follow as appendices.

## 2 Related work

In this chapter, we provide an overview of the literature that related to this thesis. Firstly, we start with an overview of requirements engineering. Later, we describe the role of AI ethical guidelines and the important quality requirements of AI systems. Next, we review the importance and challenges of developing explainable AI systems. Lastly, we provide an overview of the requirements covered in the EU AI Act.

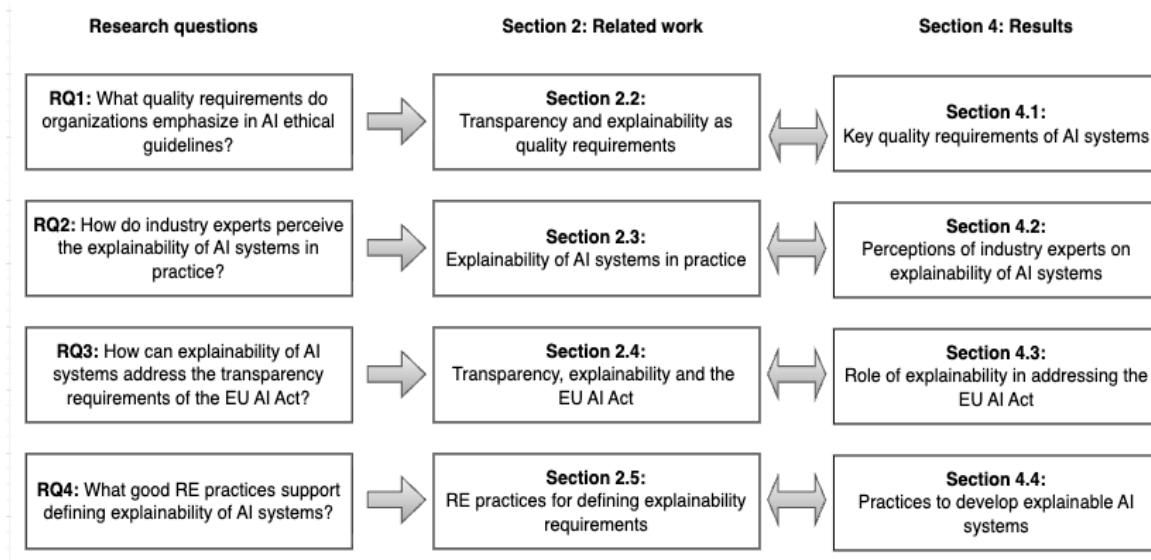


Figure 1. Structure of Section 2 and its relation to research questions and Section 4

This thesis employed an iterative literature review, conducted alongside the case studies. As Patton (1990) notes, reviewing literature throughout a study, rather than only before or after the fieldwork, helps create a creative connection between data collection and existing knowledge. This reduces bias that an early detailed literature review might introduce. In this thesis, the empirical studies on AI ethical guidelines and the definition of explainability in AI systems used the existing literature to extend the knowledge base, close the research gap, and refine the emerging findings.

## 2.1 Overview of requirements engineering

In following sections, we briefly describe the requirements engineering process, types of requirements, and established RE practices in software development.

### 2.1.1 RE activities and types of requirements

The definition of RE has evolved overtime. Zave (1997) described RE as a branch of software engineering concerned with the real-world goals, functions, and constraints of software systems. Sommerville and Sawyer (1997) offered an alternative definition, describing RE as the activity of discovering, documenting, and maintaining a set of system requirements. Furthermore, Nuseibeh and Easterbrook (2000) characterized RE as a process of identifying stakeholder needs, documenting them, and then analyzing, communicating, and implementing them. More recent perspectives emphasize customer value creation, where the values in user or customer processes are analyzed (Almquist et al., 2016). Over time, RE has shifted from a narrow focus on specification writing to becoming a foundational activity within the broader software development process (Pohl, 2010).

**RE Activities:** RE is typically carried out as a set of iterative activities aimed at capturing and satisfying stakeholder needs. Sommerville and Sawyer (1997) identify four key activities: requirements elicitation, analysis, specification or representation, and validation, complemented by the ongoing activity of requirements management.

**Requirements elicitation** is generally the starting point of the RE process (Kotonya and Sommerville, 1998; Loucopoulos and Karakostas, 1995), during which requirements are discovered by engaging key stakeholders, including customers, users, and third-party regulators (Svensson, 2011). Common elicitation techniques include interviews, ethnography, focus groups, brainstorming, and prototyping (Paetsch et al., 2003; Svensson, 2011).

Requirements analysis involves reasoning about the problem domain in order to synthesize a solution that satisfies user needs (Loucopoulos and Karakostas, 1995). During analysis, requirements are examined with stakeholder for feasibility, and conflicting or inconsistent requirements are identified and resolved. Common techniques are requirements prioritization and modelling (Paetsch et al., 2003).

**Requirements specification**, also referred to as **representation**, involves recording elicited functional and quality requirements detail (Svensson, 2011). Requirements can be represented and documented various formats, including use cases, user stories, or requirements models.

**Requirements validation** is the process of verifying that requirements are consistent, complete, and aligned with stakeholder needs (Sommerville and Sawyer, 1997). Loucopoulos and Karakostas (1995) describe validation as involving both the preparation of validation activities and the analysis of their results.



Common techniques include reviews, inspections, and checklists to detect defects, omissions, and inconsistencies (Svensson, 2011).

Finally, **requirements management** is an ongoing process of understanding, reviewing, controlling, and communicating changes to system requirements during the different development stages (Svensson, 2011). Requirements traceability, which maintains links between dependent requirements is considered a key part of requirements management (Sommerville, 2010; Svensson, 2011).

**Types of requirements:** Requirements describe the properties a system should exhibit and the needs it should fulfill (Sommerville and Sawyer, 1997; Ross and Schoman, 1977). Requirements are typically classified into two categories: functional requirements and non-functional requirements (NFRs). **Functional requirements** specify what a system should do, describing its intended behavior in terms of tasks, features, or services, such as user interface or database requirements (Sommerville and Sawyer, 1997). Functional requirements are usually collected and analyzed early in the development process.

**Non-functional requirements**, by contrast, specify constraints on how functional requirements should be implemented and are closely tied to overall software quality (Sommerville and Sawyer, 1997). NFRs span a broad range of domain-specific categories, including security, performance, usability, portability, and interoperability (Chung et al., 2009). A well-documented challenge with NFRs is that they are often not clearly identified until later stages of development, mainly due to insufficient documentation during early stages of development (Huang et al., 2006). NFRs are also commonly referred to as **quality requirements**, terminology used throughout this thesis.

### 2.1.2 Quality requirements

#### Defining and classifying quality requirements

Non-functional requirements are commonly referred to as quality requirements (Glinz, 2007; Sommerville 2010). Several definitions of quality requirements exist in the literature. Sommerville (2010) describes quality requirements as relating specific system functions, but to emergent system properties, such as reliability and response time. Thayer and Dorfman (1990) similarly define a quality requirement as one that specifies not what a system will do, but how it will do it, for example, through performance requirements. Mylopoulos et al. (1992) describe quality requirements more broadly as global requirement concerning development cost, performance, reliability, maintainability, portability, and robustness, while also noting the lack of a single, formal definition. Kotonya and Sommerville (1998) frame quality requirements as restrictions on both the product and development process. Lausen (2002) simply defines them as specifications of how well a system

performs its intended function. After synthesizing thirteen existing definitions, Glinz (2007) proposes the most concise definition: ***a non-functional requirement is an attribute of, or constraint on, a system.***

Several standards and models have been developed to classify quality requirements, including ISO/IEC 9126, McCall's model, and FURPS. The ISO/IEC 9126 model organizes quality requirements into six main characteristics, functionality, reliability, efficiency, maintainability, portability, and usability.

Chung et al. (2009) proposed a goal-oriented approach framework that organizes quality requirements into five categories: interface requirements (e.g., user-friendliness), performance requirements (e.g., time, space, and security constraints), operating requirements (e.g., physical and accessibility constraints), lifecycle requirements (e.g., design quality and development limitations), and economic or political requirements. The FURPS model, developed by Hewlett-Packard, organizes quality requirements under five root categories: functionality, usability, reliability, performance, and supportability (Chung et al., 2000). In practice, organizations often adapt or combine such models based on their specific domain and system requirements. Mairiza et al. (2010) identified 114 distinct quality requirements referenced across the literature, narrowing these down to which quality attributes are most consistently prioritized in practice.

### **Importance of defining quality requirements**

Quality requirements are widely regarded as critical to the success of software systems (Mairiza et al., 2010). Although functional requirements often receive primary attention, a system's functions provide limited value if its underlying quality characteristics are inadequate. (Chung et al., 2000). Neglecting or compromising quality requirements can lead to several negative outcomes, such as systems that are difficult to sue or costly to maintain (Ebert, 1998), reduced customer satisfaction, longer time-to-market, increased development costs, and in more severe cases, outright project failure (Svensson, 2011).

Conversely, well-defined quality requirements are closely linked to market success, since satisfying customer expectations benefits both the developing organization and its customers (Svensson, 2011). Moreover, quality requirements are also known to interact directly with functional requirements through dynamic interdependencies (Karlsson et al., 2007), further reinforcing the importance of addressing them early, ideally during the initial stages of development, as one of the most critical tasks for ensuring a system's ultimate success.

## **2.2 Transparency and explainability as quality requirements**

In this section, an overview of ethical issues of AI and ethical guidelines for developing AI system by three expert groups are summarized. Then, the quality requirements of AI systems and its link with ethical issues are outlined.

### **2.2.1 Ethical issues of AI and ethical guidelines for developing AI systems**

AI and machine learning (ML) adoption raises several interrelated ethical issues affecting users (Vampley et al., 2018), commonly categorized as autonomy (Etzioni and Etzioni, 2016, Jones et al., 2015, Palm, 2009, Royakkers et al., 2018), anonymity (Doyle and Veranas, 2014), fairness (Jones et al., 2015, Royakkers et al., 2018), privacy (Etzioni and Etzioni, 2016, Jones et al., 2015, Royakkers et al., 2018), safety (Arnold et al., 2018, Bostrom and Yudkowsky, 2011, Pieters, 2011, Jones et al., 2015), security (Jones et al., 2015, Palm, 2009, Pieters, 2011), transparency (Elia, 2009, Jones et al., 2015, Royakkers et al., 2018), and trust (Elia, 2009, Jones et al., 2015, Pieters, 2011). Autonomy, anonymity, and privacy concerns stem largely from reliance on personal data, misuse risk, and reduced individual control while fairness issue arise from data bias and discrimination that undermine value like justice and dignity (Jones et al., 2015, Royakkers et al., 2018). Moreover, growing device connectivity increases security and safety risks in critical domains such as autonomous vehicles and tracking technologies (Bostrom and Yudkowsky, 2011, Cysneiros et al., 2018, Jones et al., 2015, Royakkers et al., 2018).

There are more than 80 existing AI ethical guidelines document (Jobin et al., 2019), we reviewed three prominent frameworks: EU'S ethical guidelines for trustworthy AI (2020), IEEE's ethically aligned design (2019), SIIA's ethical principles for AI and data analytics (2017). These guidelines address issues such as transparency, autonomy, fairness, safety, and privacy, alongside societal and environmental well-being, sustainability, and risk mitigation. In addition, explainability, accountability, responsibility, and effectiveness were also covered in the guidelines. They also emphasize the need of diverse, competent teams to ensure the effective AI operation. Complementing these guidelines, Floridi et al., (2018) proposed an ethical framework centered on beneficence, non-maleficence, autonomy, justice, and explicability to support a "Good AI Society". In addition, practitioners use tools, such as data ethics canvas (Open data institute, 2019) and ethics matrix (Mephram et al., 2006).

### **2.2.2 Quality requirements of AI systems**

Compared to traditional software quality requirements, the study of quality requirements specific to AI and autonomous systems is relatively new, though

several relevant quality requirements have begun to emerge in the literature (Cysneiros et al., 2018; Leite and Cappelli, 2010).

Accountability has been proposed as an ethical value with growing relevance as a quality requirement, addressing the fundamental question of who should be held responsible for harm caused by software systems (Aydemir and Dalpiaz, 2018). Privacy, already established as a key quality requirement in traditional software, takes on additional dimensions in AI systems, encompassing personal autonomy, control over personal information, and concerns around surveillance (Cysneiros et al., 2018). Reliability remains essential for user acceptance of AI technologies, requiring systems to robustly handle real-world condition such as traffic flow and pedestrian movement. Safety plays a critical role in technology adoption, requiring rigorous testing and compliance with legal and liability standards prior to development, while security address the specific risks and consequences unique to AI systems (Cysneriros et al., 2018).

Transparency has been proposed as a particularly critical quality requirement for AI and autonomous system, enabling people to understand and interpret a system's decision and thereby fostering user confidence (Cysneiros et al., 2018; Leite and Cappelli, 2010). This includes clearly conveying what information a system used, and why and how it was used to reach a decision (Werneck, 2009). Trust is closely related, as open and transparent system behavior helps build human trust (Cysneiros et al., 2018). Furthermore, explainability has recently been proposed as distinct quality requirement for AI systems, closely related to transparency (Chazette et al., 2021). Köhl et al. (2019) argue that explainability deserves importance as a first-class requirement because AI system's increasing complexity makes their decisions difficult for stakeholders to understand. Chazette et al. (2021) similarly emphasize that explainability supports transparency and, in turn, positively influences other quality requirements such as trustworthiness and user acceptance.

### **2.2.3 Linking ethical issues and quality requirements**

Comparing the ethical issues associated with AI systems, such as transparency, security, safety, and privacy, with established software quality requirements reveals substantial overlap. Whole security, safety, and privacy have long been recognized as core quality requirements in software engineering. Transparency and trust represent newer additions, emerging simultaneous as central ethical concerns in the AI ethics literature (Jobin et al., 2019; Floridi and Cowls, 2019) and as proposes quality requirements for AI and autonomous systems specifically (Cysneiros et al., 2018; Leite and Cappelli, 2009). Jobin et al. (2019) reviewed global AI ethics guidelines and found that transparency is the most frequently cited ethical principle

across policy and industry documents. This supports the emerging concern that transparency spans both ethics and software quality.

Since the landscape of quality requirements continuous to evolve (Glinz, 2007), this overlap suggests transparency is a strong candidate as a key quality requirement in AI system. Accountability shows a similar pattern, framed as both ethical and a quality concern (Winfield and Jirotko, 2018; Aydemir and Dalpiaz, 2018). However, the importance of each requirement may vary across projects and domains, so requirements engineers must use their judgement when prioritizing them (Cysneiros et al., 2018).

This overlap between ethical concerns and quality requirements motivates the focus of this thesis. Requirements engineering already has decades of experience in eliciting, classifying, and managing quality requirements. This makes it well suited to support the systematic definition of the quality requirements of AI systems. This view aligns with recent work arguing that ethical AI principles must be put into practice through concrete techniques and methods, not left as abstract guidelines (Vakkuri et al., 2020; Ayling and Chapman, 2022). Rather than treating explainability and transparency as entirely new challenge, this thesis builds on the established foundation of requirements engineering, quality requirements.

#### **2.2.4 Transparency and explainability as quality requirements**

Guizzardi et al., (2020) define ethical requirements of AI systems as requirements derived from ethical principles or ethical codes (norms). They emphasize that defining such requirements early in the AI development process helps address ethical considerations from the early phases. Moreover, ethical requirements encompass both functional and quality requirements derived from stakeholders needs and aligned with ethical principles (Guizzardi et al., 2020, Paech and Schneider, 2020).

Transparency has been recognized as an important non-functional requirement (NFR), with early work by Cysneiros (2013) and Leite and Cappeli (2010) where they mapped the interdependencies with other NFRs, such as trust, privacy, security, and accuracy, using softgoal interdependence graphs (SIGs). Later studies emphasized transparency's particular relevance to trust, especially in domains like self-driving cars (Chazette, 2019, Cysneiros et al., 2018, Kwan et al., 2021). Some recent research has extended this focus to machine learning and autonomous systems. These studies highlight transparency as essential, yet difficult to define, because of the black-box nature (Cysneriros et al., 2020, Cysneiros et al., 2018, Habibullah and Horkoff, 2021, Horkoff, 2019). Research work by Horkoff (2019) and Chazette et al., (2019) emphasize the need for conceptual clarity and better supporting mechanisms. In response, explainability has emerged as a related

quality requirement, closely related to transparency and trust (Chazette et al., 2021, 2020, 2019, Köhl et al., 2019).

Köhl et al., (2019) proposed definitions addressing who explanations are for, when a system is explainable, and how to define explainability requirements. Chazette et al., (2021) built on this work, sharing core variables such as system, addressee, aspect, context, and added explainer as a new variable. The work linked explainability and transparency, noting that end-users mainly seek explanations during adverse situations rather than to understand internal system workings (Chazette et al., 2020, 2021). Studies also identified trade-offs between explainability and other quality requirements (Chazette et al., 2021, 2020, Köhl et al., 2019). Chazette et al. (2021) reported that explainability positively impacts qualities such as transparency, reliability, accountability, fairness, and trustworthiness, though research on integrating it into software development is still a developing research area.

The following definitions of explainability requirement and explainability are proposed by Köhl et al. (2019) and Chazette et al. (2021) respectively. These definitions served as a foundation for this thesis research work.

*“A system  $S$  must be explainable for target group  $G$  in context  $C$  with respect to aspect  $Y$  of explanandum  $X$ ” (Köhl et al., 2019)*

*“A system  $S$  is explainable with respect to an aspect  $X$  of  $S$  relative to an addressee  $A$  in context  $C$  if and only if there is an entity  $E$  (the explainer) who, by giving a corpus of information  $I$  (the explanation of  $X$ ), enable  $A$  to understand  $X$  of  $S$  in  $C$ ” (Chazette et al., 2021)*

## **2.3 Explainability of AI systems in practice**

In this section, the research literature on how explainability of AI systems is handled in practice is reviewed. Next, research on the importance of explainability is reviewed. Finally, the relevant literature highlighting the challenges of explainability of AI systems is discussed.

### **2.3.1 Explainability in practice**

In RE for machine learning, Vogelsang and Borg (2019) interviewed data scientists and emphasized the importance of eliciting explainability requirements from users' perspectives. Habibullah et al., (2023) conducted interviews and survey, but found that the importance of explainability varies by practitioner background and background, with software developers in their study not prioritizing it as a key

quality requirement. In addition, Hoffmann et al. (2023) found that over half of the professionals they interviewed expressed interest in receiving explanations.

Habiba et al. (2025) interviewed ML practitioners and identified legal requirements, such as GDPR, as key motivators for explainability, while also echoing concerns about the lack of standard practices for eliciting end-users' needs. Droste et al. (2024) performed a survey and found that explainability requirements can vary by system, underscoring the need to carefully consider them during the RE process. Next, Brennen (2020) interviewed diverse stakeholders and identified inconsistent terminology as a barrier to XAI discussion in multidisciplinary teams.

In the XAI field, Dhanorkar et al., (2021) interviewed practitioners to understand stakeholders' information needs explanation concerns throughout the AI model lifecycle and emphasized the value of a sociotechnical perspective. Liao et al., (2020) interviewed UX and design practitioners and introduced a question bank organized around the themes such as input, output, and system reasoning to support user-centered XAI development. Together, these studies primarily focus on capturing explainability needs and understanding their relevance in practice. However, further research is needed to clarify what information should be explained to users and how such explanations should be designed and communicated.

### **2.3.2 Importance of explainability of AI systems**

Studies (Habiba et al., 2025, Brennen, 2020) show that explainability improves transparency in complex ML models. Habiba et al. (2025) found that practitioners often link transparency with trust. Other studies (Vogelsang & Borg, 2019, Habiba et al., 2025, Dhanorkar et al., 2021) show that explainability helps stakeholders understand AI decision-making and internal system workings. For end-users, explanations clarify how a system reached a decision. For developers, explanations support model validation and debugging.

Studies also found that explaining model attributes and data use helps non-technical users understand what happens inside the model (Habiba et al., 2025, Dhanorkar et al., 2021). Together, these studies suggest that explainability makes AI models more insightful and understandable.

Explainability also support bias mitigation. Studies (Habiba et al., 2025, Brennen, 2020) found this to be a key reason for incorporating explainability. Brennen (2020) reported that explaining the context of the AI-generated results helps identify bias and unlawful discrimination. A study from XAI field (Dhanorkar et al., 2021) found that explainability helps clarify data characteristics, such as whether data is public, proprietary, or used for training. This clarity helps stakeholders validate data sources and discuss data use and reuse with clients.

A healthcare study (Sporsem et al., 2026) identified that explainability helped clinicians evaluate AI predictions against their own judgement. However, the same study found that users did not need detailed explanations of the AI system's internal workings.

### **2.3.3 Challenges of explainability of AI systems**

Although many studies highlight the importance of explainability, four studies identify specific challenges to its adoption in practice. Two studies (Habiba et al. 2025, Liao et al. 2020) report difficulty balancing explainability with other quality and legal requirements. For example, understandability and accuracy of explanations can be hard to balance (Habiba et al. 2025). Complex and high-performance models are also often less interpretable, creating trade-off between performance and explainability. Explainability can further conflict with safety and legal requirements, particularly around sensitivity (Habiba et al. 2025) and algorithmic confidentiality (Habiba et al. 2025, Liao et al. 2020, Dhanorkar et al. 2021). Dhanorkar et al. (2021) highlights that deciding how much to explain, without revealing too much about the technology, is an often-overlooked challenge tied to confidentiality and safety.

Communication challenges also arise from knowledge gaps between technical and non-technical stakeholders. Practitioners report struggling to understand users' background, domain knowledge, and explainability needs (Habiba et al. 2025). Similar knowledge gaps exist between development teams, particularly between designers, testers, and AI/ML practitioners (Liao et al. 2020, Dhanorkar et al. 2021), with Dhanorkar et al. (2021) noting a steep learning curve for teams building explainable AI systems.

Resource constraints present another major challenge. Habiba et al. (2025) point to computational resource limits when scaling explainability in larger systems, while Liao et al. (2020) highlight time and resource costs as concern for product teams. Maintaining explainability is also difficult, since changing data often requires model retraining (Habiba et al. 2025).

Building users trust remains challenging, as explaining how results are generated is rarely straightforward (Habiba et al. 2025), a difficulty closely related to the complexity of black-box models (Habiba et al. 2025, Liao et al. 2020). Practitioners also point the lack of standardized approaches and best practices (Habiba et al. 2025). In the XAI field, disagreement between post hoc explanation methods is a notable challenge. Krishna et al., (2024) found that when explanations from tools like LIME, KernalShap, Vanilla Gradient, and SmoothGrad conflicted, practitioners often relied on intuition or were unsure how to proceed, revealing a lack of structured approaches for resolving such disagreements.



## **2.4 Transparency, Explainability and the EU AI Act**

The section is divided into three subsections. First, an overview of the EU AI Act is described. Second and third, the existing literature relating to explainability and the AI Act, and transparency and the AI Act are reviewed.

### **2.4.1 Transparency requirements in the AI Act**

The EU AI Act (2024) is a harmonized legal framework regulating the development, deployment, and use of AI systems across the EU, aiming to protect fundamental rights, safety, and democracy while supporting the free movement of trustworthy, human-centric AI (Recitals 1,2,8). The act sets requirements for high-risk AI systems, including transparency, human oversight, accuracy, and robustness (Article 1), while also safeguarding innovations for SMEs and startups (Recital 25). Figure 1 summarizes the requirements highlighted in the AI Act (EU AI Act, 2024).

Transparency is a core requirement for high-risk systems (Article 13), building on the European commission's Ethics Guidelines for Trustworthy AI (2019), which frame transparency as encompassing explainability, traceability, and open communication about AI use. Traceability supports auditability and explainability by documenting data, algorithms, and decisions, while explainability, in the ethics guidelines, refers to an AI system's ability to explain both its technical processes and human decisions.

The AI Act introduces instructions for use (Recitals 72, 91; Articles 13,14,26) to operationalize these concepts for providers and deployers, through explainability itself appears only implicitly across several recitals and articles, linked to several quality requirements such as fairness, accuracy, robustness, cybersecurity, and trustworthiness.

The AI Act's communication requirement support risk management and apply to providers, deployers, and regulatory bodies. Providers must transparently communicate an AI system's capabilities and limitations to deployers (Recital 27). They must also supply documentation explaining technical elements relevant to decision-making (Article 13). Involving external experts is recommended for high-risk systems (Recital 65). Providers should also give narrative explanations of training data sources (Recitals 107).

For deployers, the AI Act emphasizes user awareness of the right to explanation (Recital 93). For high-risk systems, deployers must provide clear explanations of the AI system's role, its decision-making process, and the key factors behind its decision (Recital 86). Finally, the Act requires the European Commission and national authorities to create AI sandboxes. These sandboxes let providers develop and test tools for explaining accuracy, robustness, and cybersecurity (Article 58).

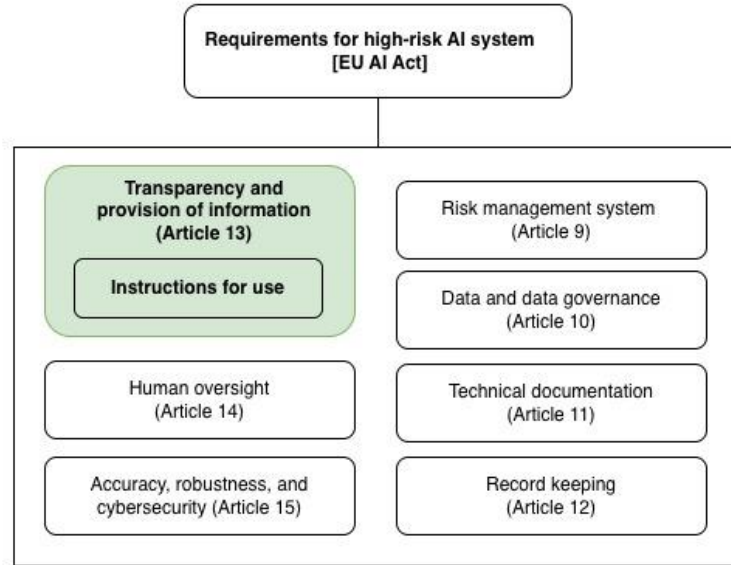


Figure 2. Overview of the EU AI Act requirements (Paper IV)

#### 2.4.2 Explainability and EU AI Act

Recent research on explainability in connection to the AI Act highlights explainability is essential for transparency, accountability, and trustworthy AI governance. Panigutti et al. (2023) indicated the contrast between the AI Act regulation and existing scientific research, where the former mandates capturing AI system requirements, while the latter majorly focuses on using understandable models used in the AI system. A key takeaway from their study is that AI Act does not mandate using XAI tools or a transparency-by-design approach when building AI systems. Instead, it emphasizes obligations regarding risk management systems, transparency requirements, and human oversight in the AI Act Articles 9, 13, 14 respectively (Panigutti et al. 2023).

Pavlidis et al. (2024) emphasizes XAI as a prerequisite for effective oversight and regulation of high-risk AI systems. Additionally, Nannini (2024) analyze challenges of implementing the right to explanation in the AI Act and proposes a framework to address the legal, technical, and socio-ethical tensions. Further, Dhaini et al. (2025) in their study focusing on implementing XAI to comply with EU AI Act identified technical complexity, regulatory vagueness, and accuracy trade-offs as key challenges.

Moreover, Kelly et al. (2024) study on mapping AI Act requirements to quality attributes indicated that explainability is a significant quality requirement. The study correlated explainability with Article 14 (Transparency and provision of information) and Article 15 (Human Oversight) of the AI Act and provides examples of documentation requirements of explainability that need to be ensured by the AI service providers when designing AI systems.

### 2.4.3 Transparency and EU AI Act

Currently, the research on critically examining the notion of transparency within the EU AI Act is growing. Varošaneć (2022) provides a systematic overview of transparency provisions in the AI Act. Their work proposes inclusive co-regulation, rather than self-regulation to better protect both innovators and individuals affected by AI decisions. Busuioc et al. (2023) take a more critical view and argue that the Act frames transparency as mediated explanation, largely controlled by AI providers. In addition, they specify that the Act has shaped transparency away from its original meaning as open disclosure, creating opportunities for opacity and weakening accountability. Moreover, the work (Busuioc et al., 2023) highlights that the transparency should be reclaimed as genuine disclosure, not provider-controlled explanation. Gyevar et al. (2023) identify a “transparency gap” between the AI Act and the field of XAI. The act treats transparency as means to support broader values, such as accountability and human rights. In contrast, XAI treats transparency narrowly as an end. The work proposes four areas where this gap could be addressed, including clarifying the scope of transparency and building explainability for datasets (Gyevar et al. 2023).

Other research focuses on how transparency requirements are implemented and enforced in practice. Söderlund (2024) examines the transparency mandates effectively to create a model of “qualified transparency”, which may help reduce AI opacity. However, the work notes that significant challenges remain in meeting the Act’s broader goals. Genoveri et al. (2024) address definitional inconsistency more directly. They compare transparency definitions and requirements across several AI standards and regulations, including the AI Act. Most documents address transparency and explainability, but specific requirements, such as what information must be provided, to whom, and in what context, vary considerably and remain vague. This points to a clear need for standardization.

Finally, more applied research addresses the technical and economic sides of these requirements. Kalodanis et al. (2024) examine the challenge of achieving AI-Act compliant transparency in large language models (LLMs), given their black-box nature. The work review explanation techniques, from attention mechanisms to post hoc explanation tools, and propose guidelines covering performance-interpretability tradeoffs, documentation and user-centered design. From an economic perspective, Holst et al. (2023) conducted a multivocal literature review on the AI Act’s impact on AI innovation. The study found that transparency requirements do not necessarily hinder the patentability of AI innovations. Instead, most concerns stem from uncertainty around key terms in the Act and hints the necessity to resolve the ambiguity. Overall, this research shows that transparency is a foundational value in the AI Act, but considerable ambiguity remains in how it is defined, applied, and enforced in practice (Holst et al. 2023).

## 2.5 RE practices for defining explainability requirements

A systematic literature review conducted as part of this thesis identified a range of existing RE practices that support the definition, design, and evaluation of explainability requirements of AI systems (Paper III). The RE practices that support in defining the explainability requirements based on the literature review are presented in Table 3.

Four practices were found to serve as prerequisites for defining explainability requirements: identifying stakeholders and their explainability needs (Nguyen et al., 2021; Chazette et al., 2020), understanding organizational business goals related to explainability (Chazette et al., 2022), establishing a shared understanding of users and organizational processes to provide context, and clearly defining the AI system's purpose from the perspectives of both users and other stakeholders (Philips et al., 2021).

Several established elicitation techniques were also found applicable to explainability requirements. Interviews are widely used to uncover users' explanation needs and expectations, and to analyze when and how explanations should be designed (Nguyen et al., 2021; Chazette et al., 2020; Chazette et al., 2022). User observation, task analysis, and ethnographic studies similarly support the identification of explanation needs and deepen understanding of the explainability context and domain (Chazette et al., 2020; Chazette et al., 2022).

Questionnaires were found useful across multiple RE phases, supporting elicitation, validation, and evaluation of users' explainability needs and perceptions (Ramos et al., 2021; Nguyen et al., 2021; Chazette et al., 2020). Workshops, brainstorming sessions, and multidisciplinary teams were repeatedly highlighted for clarifying technical business, and environmental aspects of explainability, with multidisciplinary workshops in particular found to combine both organizational and system level perspectives (Chazette et al., 2022). Brainstorming was highlighted as useful for shaping explainability approaches and improving team communication when explaining algorithms. In addition, focus groups were also identified as relevant during both elicitation and evaluation (Chazette et al., 2022).

Beyond elicitation, several practices support requirements analysis and modeling. Requirements catalogue help identify conflicts between explainability and other quality requirements, such as privacy, security, and usability. Drawing on prior work such as Chazette et al.'s extensive explainability requirements catalogue (Köhl et al., 2019; Chazette et al., 2020, 2021, 2022) and can further support the construction of soft-goal interdependency graphs (SIGs) which document the reasoning behind explainability design decisions and clarifying interrelationships and conflicts among quality requirements (Köhl et al., 2019; Li & Han, 2023).

Table 3. Overview of practices for defining explainability requirements (Paper III)

| RE activities  | RE practices                                                                                                                                                                                                                                                                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prerequisites  | <ul style="list-style-type: none"> <li>• Identifying stakeholder goals</li> <li>• Identifying business goals</li> <li>• Identifying users' and organizational process</li> <li>• Defining purpose of AI system</li> </ul>                                                                                                                         |
| Elicitation    | <ul style="list-style-type: none"> <li>• Identifying explainability needs</li> <li>• Interviews</li> <li>• User observation</li> <li>• Questionnaires</li> <li>• Task analysis</li> <li>• Ethnographic studies</li> <li>• Workshops</li> <li>• Brainstorming sessions</li> <li>• Focus groups</li> <li>• Using multidisciplinary teams</li> </ul> |
| Analysis       | <ul style="list-style-type: none"> <li>• Requirements catalogue</li> <li>• Interviews</li> <li>• Analyzing potential risks and negative consequences</li> <li>• Workshop</li> <li>• Brainstorming sessions</li> </ul>                                                                                                                             |
| Representation | <ul style="list-style-type: none"> <li>• Empathy maps</li> <li>• Personas</li> <li>• Templates</li> <li>• Soft-goal interdependency graphs (SIGs)</li> </ul>                                                                                                                                                                                      |
| Validation     | <ul style="list-style-type: none"> <li>• Negotiating requirement tradeoffs</li> <li>• Identifying non-feasible requirements</li> <li>• Questionnaires</li> </ul>                                                                                                                                                                                  |

Analyzing potential risks and negative consequences was found useful for identifying stakeholders' concerns and negative perceptions of AI use during requirements analysis (Philips et al., 2021). Furthermore, empathy maps and personas support the modeling of explainability needs from an emotional and user-centered perspectives (Ramos et al., 2021, Chazette et al., 2022). Structured templates, including Volere-based formats were also proposed to consistently represent individual explainability requirements (Nguyen et al., 2021; Chazette et al., 2022). Finally, negotiating requirements tradeoffs and identifying infeasible requirements were identified as important for validating explainability requirements under conditions of uncertainty (Habiba et al., 2022; Philips et al., 2021), with requirements catalogues further supporting this tradeoff analysis (Köhl et al., 2019).

## 3 Research Method

This thesis is an exploratory study to understand the role of requirements when developing explainability of AI systems. In this section, the goals of the research and the respective research questions are described in detail. As the main goals of the thesis are to bridge the gap in literature, to reveal the current state in practice, and to understand human perspectives, the work is exploratory and follows socio-technical grounded theory relating to subjectivist epistemology and constructivist approach (Hoda, 2024). To study the current state in practice, we utilized case study as the primary research method combined with qualitative methods in empirical software engineering (Seaman, 1999). Moreover, it is critical to transfer knowledge from academia to industry and for this process we used the emerging technology transfer research method (TTRM) which is described in Section 3.2. The research methods we used are described in the following subsections.

### 3.1 Overview of the Research Methods

The high-level research problem that this thesis aims to address is: *How requirements engineering support defining explainability of AI systems?*

The following research questions can help understanding the current state of explainability in practice and identify how requirements engineering can support defining explainability of AI systems.

- RQ1: What quality requirements do organizations emphasize in AI ethical guidelines?
- RQ2: How do industry experts perceive the explainability of AI systems in practice?
- RQ3: How can explainability of AI systems address the transparency requirements of the EU AI Act?
- RQ4: What good RE practices can support defining explainability of AI systems?

Furthermore, Table 3 provides an overview of research methodologies used in our publications, along with the data collection methods.

Table 3. Publications with their research methods

| <b>Publication</b> | <b>Research Method</b>                                                 | <b>Data Source</b>                  | <b>Data analysis</b>                                          |
|--------------------|------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------|
| I                  | Case study                                                             | Interviews and Documents            | Within case analysis and cross-case analysis                  |
| II                 | Qualitative approach & Technology transfer research methodology (TTRM) | Documents, Interviews, and Workshop | Conceptual ordering, line-by-line coding, word-by-word coding |
| III                | Literature review                                                      | Information from literature         | Thematic analysis                                             |
| IV                 | Case study                                                             | Interviews, Documents, and Workshop | Thematic analysis (Deductive and inductive analysis)          |

### 3.2 Qualitative studies

We conducted our empirical studies of this thesis using qualitative methods, including interviews, document analysis, and workshops (Paper I, II, and IV). For Paper I, the multiple case study (Yin, 2013, Cassell and Symon, 2012) enabled us to examine the current state of AI ethics and ethics guidelines in companies. In Paper II, Phase 1 expanded the scope of Paper I by examining AI ethical guidelines across 16 organizations, in phase 2, we used the technology transfer research methodology (TTRM) which supports industry-academia collaboration by evaluating the research solutions in real-world projects (Wohlin and Runeson, 2021). Finally, in Paper IV, we conducted a case study following the approach proposed by Runeson et al. (2012) to study the current state of explainability in practice.

#### 3.2.1 Case studies

The two case studies in this thesis are a multiple case study (Paper I) and a single case study (Paper IV) aiming to study the current state of ethical AI development in practice and the current state of explainability in practice respectively. For the multiple case study, the methodology was designed according to Yin (2013), and, in the single case study, it was designed according to Runeson et al. (2012). A detailed description of the case organization including the size of the company, reasons for selection, participants role in the organization can be found in Paper I, II, and IV.

#### *Case organizations*

For the multiple case study, we selected three organizations that have invested in the development of ethical AI. The first organization is software consulting company that designs and delivers AI services and products and had actively organized ethics coaching for their employees and customers. The second organization is a Finnish retail company involved in car, food, and building trades and have AI ethics discussion internally in their teams. The third organization is a finance service provider with millions of customers and has dedicated AI ethics team since 2017.

For our single case study, we selected a consulting company operating in Finland, Norway, Sweden, and six other European countries. The organization strives to develop responsible and sustainable AI and has made a significant investment in GenAI. We selected this organization for two reasons: 1) they develop AI systems in different domains such as public sector, healthcare, and finance, 2) the organization collaborates with universities to explore explainable AI.

### *Interviews*

We conducted three interviews as a part of our multiple case study. An AI expert who has knowledge about the case organizations. The interviewees were deeply knowledgeable about the ethical guidelines of their companies and had closely collaborated with professionals of various backgrounds such as data scientists, designers, and developers.

For our single case study, we conducted eight interviews. Our goal was to interview industry experts who are involved in AI development. We used purposive sampling to involve industry experts from different roles to understand the current state of AI development and explainability. The industry experts had expertise in areas such as technology, social, design, business, and ethics.

### *Documents*

The case studies of this thesis also used document analysis technique along with the interviews to ensure data triangulation. For the multiple case study, we collected two documents from each case company, one is their AI ethical guidelines document and the other is an internal document or training document they have for related to ethical AI development. For our single case study, we collected six relevant documents that covered topics such as sustainable AI, AI Act, and trustworthy AI. The summary of these documents can be found in the Paper IV.

### *Workshop*

For Paper IV, we conducted a workshop to present the preliminary findings and discuss them. The workshop was attended by five industry experts whom we also interviewed as a part of the case study. We used proposals of Creswell and Miller (2000) and McKim (2023) for member-checking to design the workshop session.



The participants were asked to share their general thoughts on the preliminary findings, discussed the important findings, and findings that differed from their experience.

### *Data collection and analysis*

In the multiple case study, the interview questions were designed following the guidelines of Boyce and Neale (2006) and the questions were improved based on the feedback of three senior researchers. Two pilot interviews were conducted to ensure its feasibility and understandability. The lengths of the interviews were from 60-80 minutes.

We analyzed the data from the interviews and documents using the Eisenhardt (1989) research method. First, we performed a within case analysis and created a case description report for each company by categorizing the codes based on the AI ethical issues. Next, we employed cross-case analysis method during which the evidence data and codes from each case is compared to the other cases and the results were reported based on it.

In our single case study to examine the current state of explainability, we conducted intensive interviewing to explore the experiences and perspectives of industry experts, according to Charmaz (2006). As a first step, we familiarized ourselves about the organization and checked the case organization's website to identify reports, guides, examples, and blog posts published relating to the organization's AI and ML development. Then, we developed interview questions using the concepts and terminologies that are relevant to the prospective interview participants. A total of eight interviews were conducted ranging from 80-90 minutes per interview to gain deeper and richer insights.

We performed a six-phased thematic analysis proposed by Braun and Clarke (2006) and Nowell et al. (2017). In the first phase, the researchers of the study familiarized the interview data by reading the transcripts. Second, we performed initial coding and started peer briefing of the codes and emerging themes. In the third phase, the codes were collated to form themes, and interview codebook and concept maps for each theme is created. The fourth phase focused on reviewing the themes, concepts and their connections. In the fifth phase, we finalized the names of the concepts and their relationship and captured any misinterpretation in the relationships through peer debriefing. As a final step, we reported the results of thematic analysis, and the preliminary report was reviewed by all the authors of the study.

The importance of the AI Act emerged during the inductive thematic analysis. In the next iteration, we analyzed the AI Act, focusing especially on the transparency requirement to develop a lightweight approach consisting of principles, practices, and guiding questions. This lightweight approach partly answers the third research question of this thesis.

### 3.2.2 Technology transfer research methodology (TTRM)

In the empirical study of Paper II, we studied the AI ethical guidelines in 16 organizations and aimed to explore how define explainability requirements of AI systems can be defined for a real-world AI system. The study was conducted in two phases. Phase 1 focused on analyzing the AI ethical guidelines for developing transparent and explainable AI systems and propose solutions to define explainability requirements. Phase 2 focused on evaluating the solutions proposed in Phase 1.

We selected technology transfer research methodology for two reasons: 1) this research methodology highlights gradual empirical evaluations of solutions, 2) the methodology provides concrete guidelines on how these can evaluation can be performed as a industry-academia collaboration (Gorschek et al., 2006, Gorschek et al., 2020, Wohlin and Runeson, 2021). For our study, we followed the TTRM according to Wohlin and Runeson (2021) that contained six steps: 1) identify the industrial challenge, 2) Assess practice and formulate a research objective, 3) study the state-of-the-art, 4) develop one or more candidate solution(s), 5) evaluate the solution(s), and 6) move the chosen solution into practice and evaluate.

We conducted the first three activities of TTRM iteratively. First, AI ethical guidelines from three organizations were analyzed alongside practitioners' interviews and the results were reported in Paper I. Additionally, a literature review was performed on potential ethical issues of AI system and compared it with AI ethical guidelines. In the second iteration (Paper II – phase 1), we focused on transparency and explainability guidelines. We used the explainability definition Chazette et al. (2021) as a starting point when analyzing the ethical guidelines of 16 organizations.

During the fourth activity, we developed two candidate solutions, which are the model of explainability components and the template for representing explainability requirements. Next, we started with a static testing of the solutions in academic setting and then decided to conduct a small-scale pilot evaluation with practitioners for a real-world AI system. Finally, we shared our experience in defining the explainability requirements with practitioners as lessons learned and based on them, we suggested six good practices that can support in defining explainability requirements in phase 2 results of Paper II.

#### *Case organizations*

For the phase 1 study, we used the AI ethical guidelines of 16 organizations. We classified the 16 organizations into three categories: 9 professional services and software organizations, five business-to-consumer (B2C) organizations, and two public sectors.

For the phase 2, we selected a technology, data, and design company which is one of the biggest data consulting houses in the Nordic countries with around 600

data engineers and data scientists. Moreover, the organization was interested in exploring the possibilities of incorporating transparency and explainability in AI systems they develop.

#### *Case system – recruitment game AI system*

The recruitment game was designed to satisfy two goals: recommending suitable open positions to job applicants based on game scores, education, and work experience, and helping recruiters identify suitable candidates based on demonstrated skills. Applicants first complete four programming tasks within the game to earn points, after which an AI model suggests matching open position in the organization by combining the points earned, education, and experience. Recruiters, in turn, view applicants' scores, backgrounds, and suggested positions to review and select suitable candidates for available roles.

The reason for choosing the recruitment game system is that it was in its early prototype stage which is an opportunity to elicit explainability requirements and ensure explainability of the system. Hence, we applied the two candidate solutions i.e., the model of explainability components and the template for defining explainability requirements of the recruitment game system

#### *Interviews and workshop*

We conducted three interviews with practitioners who had knowledge of the recruitment game system. The interviews aimed to gather details, such as purpose of the system, who developed the game, how the game works, who developed the AI part, and how the AI works. In addition, we also aimed to capture interviewees viewpoint on explainability.

The workshop was conducted after the interviews. The workshop ensured the discussion between practitioners and researchers on their explainability perspectives. The workshop had in total of seven participants, 5 out of them were recruiters and two were the developers of the recruitment game.

#### *Data collection and analysis*

The interviews were open-ended and semi-structured, and it lasted from 45-90 minutes. One of the three interviews was held face-to-face, while the others were organized on Zoom. Next, the workshop session was planned and facilitated by three researchers. The workshop was two hours long and covered topics such as participants view on transparency and explainability, positives and negatives of using AI in recruitment, using the model of explainability components for the recruitment game system to elicit the target groups, aspects, context, and explainers of explainability.

For the data analysis, the interview transcripts were analyzed to identify key themes such as goals of the recruitment game, functionalities of the game, people

involved, data used by AI, etc. The results this analysis were useful in workshop planning. For the workshop transcript, two researchers performed word-by-word coding. The coded concrete examples of target groups, aspects, contexts, and explainers. Next, in axial coding, we clustered the examples and identified the relationships between them. A practitioner who is also a researcher involved in our study reviewed the results to make sure the data are interpreted correctly during the analysis.

### **3.3 Literature reviews**

A literature review was a part of most of the studies of this thesis. In papers I, II, and IV, a light-weight review of explainability literature from the research question viewpoint was performed. Paper III was based purely on a review of literature focusing on explainability requirements in practice.

For the literature review of paper III of this study, we did not follow any systematic literature review (SLR) guidelines. Alternatively, the literature review used an experience-based literature review approach suggested by Kuhrmann et al. (2017). The fundamental reason for selecting this approach is that this method provides practical guidance on data collection and selection processes, which is very useful for young researchers performing a comprehensive literature study. Kurhmann et al. (2017) suggest three main phases for study design. They are:

- 1) Preparation phase – The research questions, study queries, and inclusion and exclusion criteria of the study were defined.
- 2) Data collection and data cleaning phase – The data sources from where we collected our data for the literature review include prominent RE journal, RE conferences, RE workshops, and standard libraries such as IEEE Xplore, ScienceDirect, and SpringerLink. Then a set of papers were shortlisted which matched our search strings.
- 3) Study selection – The inclusion and exclusion criteria were applied on the shortlisted papers and then the final relevant set of papers were selected.

For research papers I, II, and IV, all relevant articles that were found were reviewed based on the research questions and inclusion/exclusion criteria. The reason for choosing an informal approach is to make sure to include only relevant articles in the related work section, as a general search on explainability retrieved numerous articles within the XAI field, which were out of our research scope. Moreover, relevant papers from the SE and RE fields may not have the keyword ‘explainability’ listed but could reveal some key information as part of transparency and AI ethics topics. Consequently, the goal of the light-weight literature review,

often expressed as related work, was not to draw conclusions based on existing literature, but rather, to show relevant published existing knowledge.

## 4 Summary of the results

The results of the research are summarized in this section. In the respective research articles, a detailed description of each study and their results can be found. This section is structured in accordance with the four research questions (RQ1-RQ4) of this thesis.

### 4.1 Key quality requirements of AI systems

The first research question of this thesis investigated what quality requirements are highlighted in the AI ethical guidelines of organizations. This research question is answered in Publication I and Publication II. Publication I focused on discovering what ethical issues does the organizations aim to cover in their AI ethical guidelines and their connection to defining critical quality requirements of AI systems. Publication II summarized the quality requirements that are emphasized in the AI ethical guidelines of 16 organizations. This section provides an overview of quality requirements that are important when developing AI systems.

#### 4.1.1 Overview of ethical guidelines of AI systems

The analysis of AI ethical guidelines was performed and presented Paper I and Paper II. The results indicate that organizations defined their own sets of AI ethical guidelines to handle the ethical issues of AI that are relevant and important to their organization. The organizations addressed certain quality requirements in their AI ethical guidelines.

The 16 organizations in our study were categorized into three main categories: A – professional services and software organizations (O1-O9), B – business-to-consumer (B2C) organizations (O10-O14), and C – public sector organizations (O15, O16). We identified eight key quality requirements that were repeated in the AI ethical guidelines and mapped them to the respective organizations. The following tables summarize the quality requirements emphasized in the AI ethical guidelines of 16 organizations.

Table 4. Quality requirements in ethical guidelines of Category A

| Quality Requirements | Professional services and software |    |    |    |    |    |    |    |    |
|----------------------|------------------------------------|----|----|----|----|----|----|----|----|
|                      | O1                                 | O2 | O3 | O4 | O5 | O6 | O7 | O8 | O9 |
| Transparency         | x                                  | x  | x  | x  | x  | x  | x  | x  | x  |
| Explainability       | x                                  | x  | x  | x  | x  | x  |    | x  | x  |
| Privacy              | x                                  | x  | x  | x  | x  | x  | x  |    | x  |
| Security             | x                                  |    |    |    | x  | x  |    | x  | x  |
| Safety               |                                    |    |    | x  |    | x  |    | x  |    |
| Fairness             | x                                  |    | x  | x  | x  | x  |    | x  | x  |
| Accountability       |                                    |    | x  | x  | x  |    | x  |    | x  |
| Reliability          |                                    |    |    |    | x  | x  |    |    |    |

Table 5. Quality requirements in ethical guidelines of Category B and C

| Quality Requirements | B2C |     |     |     |     | Public Sector |     |
|----------------------|-----|-----|-----|-----|-----|---------------|-----|
|                      | O10 | O11 | O12 | O13 | O14 | O15           | O16 |
| Transparency         |     | x   | x   | x   | x   | x             |     |
| Explainability       |     | x   | x   | x   | x   | x             |     |
| Privacy              |     | x   |     | x   | x   | x             | x   |
| Security             | x   | x   | x   | x   |     |               | x   |
| Safety               |     | x   |     |     |     |               | x   |
| Fairness             | x   | x   | x   | x   |     |               |     |
| Accountability       |     | x   |     |     | x   |               | x   |
| Reliability          |     |     |     |     |     |               | x   |

*Privacy* ethical guidelines across organizations emphasized protecting personal and sensitive data, avoiding its unethical use, and complying with regulations such as GDPR. B2C and public sector organizations prioritized customer confidentiality and privacy, along with adherence to data protection practices. A few of professional services, software, and B2C organizations addressed security and privacy together. It emphasized data safety, misuse prevention, risk reduction, and compliance with *safety* principles.

Organizations promote *fairness* by developing unbiased and inclusive AI systems that prevents bias and discrimination. While B2C organizations emphasize on making the decision-making process equal and fair, professional services and software organizations highlight equality, diversity, and inclusiveness as foundations of responsible AI. Next, organizations promote *accountability* by assigning human owners who is responsible for AI decisions, operations, and ethical concerns. Some organizations further support accountability by establishing audits, oversight forums, and ethics communities throughout the AI lifecycle. Professional services and software organizations focus on *reliability* through safety and quality standards that assess AI risks before deployment. Public sector organizations emphasize the use of reliable data and algorithms that matches the AI system's intended purpose.

#### 4.1.2 Transparency and explainability as quality requirements

*Transparency* was one of the widely defined ethical guidelines, with 14 of the 16 organizations addressing transparency. Notably, all professional services and software organizations defined transparency guidelines for their AI development. These guidelines primarily focused on the use of AI and communicating openly about how and where the AI systems are used. Interestingly, *explainability* is consistently integrated into transparency guidelines in 13 of the 14 organizations of our study. Our study (Paper I) suggest that organizations can use their ethical guidelines to identify and prioritize critical requirements of AI systems. Based on the importance in ethical guidelines, we indicate that transparency and explainability can be critical quality requirements of AI systems (Paper I and II).

##### *Transparency*

Across the organizations (Paper I and II), the importance of transparency was highlighted by several interrelated reasons. The most prominent reason was building and maintaining users trust, cited by nearly one third of the participating organizations of our study. Transparency was also linked to security, with a couple of organizations indicating that transparency practices support the security of AI systems. Moreover, transparency is also important for specific functional reasons, one organization emphasized that transparency helps distinguish between instances where AI systems make autonomous decisions and instances where they only support the human decision-making. Transparency also contributes to mitigating unfairness in addition to fostering user trust. Beyond these organization-specific reasons, transparency also helps in enabling broader accountability, allowing AI systems to be assessed for external review and scrutiny. Overall, transparency is driven by a combination of user-centric goals such as building trust, and broader organizational goals such as strengthening security, fairness, and accountability in AI systems.

##### *Explainability*

According to the organizations, explainability is a core and integral part of transparency, as the organizations always coupled them in their AI ethical guidelines. Similarly, the EU ethics guidelines for trustworthy AI (European commission) portrayed explainability as one of the key requirements of transparency, along with traceability and communication. The analysis of the transparency ethical guidelines helped us derive a model of explainability containing four key components. The purpose of the model is to assist the practitioners in eliciting explainability requirements of the AI system. The four components of explainability are: target groups, aspects, context, and explainers. The model represents a good practice for developing explainable AI system, which is discussed in a later section.



### *Quality requirements related to transparency and explainability*

The analysis of the AI ethical guidelines showed that transparency and explainability are closely related to several other quality requirements. Organizations consistently associated *understandability* with transparency and explainability by emphasizing that stakeholders should understand how AI systems are used, how they behave, where and why AI is applied, and whether AI makes autonomous decisions or only supports human decision-making. In addition, *traceability* is identified as an important requirement for enabling transparency by supporting tracing of AI decisions and the data used throughout the decision-making process of the AI system.

Organizations emphasized that transparency and explainability contribute to *trustworthiness* of AI systems by promoting openness about AI operations and providing explanations to those affected by AI decisions. *Privacy* emerged as a related but occasionally conflicting quality requirement, as organizations noted that aiming for transparency can create tensions with privacy considerations. *Auditability* was also linked to transparency as organizations highlighted the need to design AI systems that are ready for auditing, while transparency was further associated with ensuring *fairness*.

In summary, the results for RQ1 show that organizations can use their AI ethical guidelines to identify and prioritize critical quality requirements, with accountability, fairness, privacy, reliability, transparency, and explainability emerging as key quality requirements across the 16 organizations studied. Among these, transparency and explainability stood out as the most consistently emphasized. Both were also shown to impact several related quality requirements, including understandability, traceability, trustworthiness, and auditability, reinforcing their role as a foundational and interconnected in ethical AI development. These findings suggest that while organizations address a broad range of ethical concerns, transparency and explainability represent the most central and consistently prioritized quality requirements of AI systems.

## **4.2 Perceptions of industry experts on explainability of AI systems**

The second research question of this thesis was to understand how industry experts perceive the explainability of AI systems in practice. To answer this research question, a case study (Paper IV) was conducted to determine the importance of explainability in practice and to identify the challenges of implementing explainability in AI systems from the perspectives of industry experts. The research question was also implicitly addressed in the Paper II (Phase 2 - workshop) that focused on defining explainability requirements for their then-upcoming

recruitment game system. The workshop participants expressed interest in defining explainability requirements and perceived explainability of AI systems as beneficial.

#### **4.2.1 Reasons for explainability in practice**

From the case study, we identified the following reasons for explainability:

- to foster users' trust in AI systems
- to increase users' understanding of AI systems
- to handle risks and limitations of AI systems
- to comply with national law and the AI Act
- to ensure accountability and human agency

First, explainability fosters users' trust, as experts noted that understanding how and why an AI system behaves is essential for user acceptance and long-term adoption, particularly in domains where end-users must rely on system outputs.

Second, explainability increases the users understanding of AI systems, helping them comprehend how results are generated, the factors influencing decisions, and the limitations of AI systems, including risks such as unreliable or hallucinated results in black-box models. This understanding was also seen necessary for responsible use of AI systems, regulatory compliance, and communication among developers, users, and legal experts.

Third, explainability helps organizations communicate the risks and limitations of AI systems. Experts emphasized that users should understand the error-prone nature of ML outputs, and that publicly communicating such limitations improves organizational risk awareness and responsible development practices.

Fourth, explainability supports compliance with national law and the AI Act, as certain sectors, such as public sector, healthcare, banking, and insurance are legally required to justify AI decisions. The adoption of the AI Act further reinforced the need for explainability in high-risk systems.

Fifth, explainability was linked to accountability and human agency, particularly in public sector, where AI systems can significantly affect citizens' life. The industry experts recommended that private organizations should also establish high standards of explainability especially for AI systems that have direct impact on people such as loan approval, credit card or healthcare eligibility.

Nevertheless, the industry experts underscored that is context-dependent and user-dependent. Explainability is not essential for all AI systems, for example, automation of some menial tasks may not require explainability, whereas critical AI systems with significant consequences for users demand transparency, accountability, and trust.

#### 4.2.2 Explainability challenges of AI systems in practices

The perceived challenges of explainability in practice is summarized in the figure below.

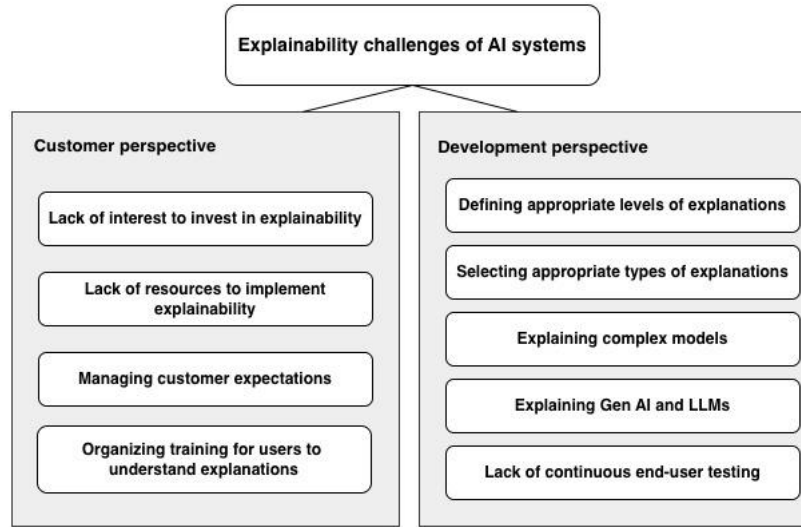


Figure 1. Explainability challenges in practice

##### *Explainability challenges – customer perspective*

From the customer perspective, four key challenges emerged. First, customers often show a lack of interest in investing in explainability, prioritizing using AI technologies in their systems over understanding how the AI is used and their impact.

Second, organizations face a lack of resources to implement explainability. The industry experts highlight insufficient education on ethical AI development, difficulty hiring qualified personnel, and limited investment in this area as reasons for this challenge in practice.

Third, managing customer expectations poses a challenge. Sometimes customers prefer cutting-edge AI in their AI solutions for marketing purpose, even when the same solutions could be developed by simpler and explainable models.

Fourth, organizing training for users to understand explanations is in some cases. Particularly, providing training for users is essential for critical use cases whose explainability needs are higher. This challenge remains closely linked to the development-side decision about explanation level and format.

##### *Explainability challenges – development perspective*

From the development perspective, five distinct challenges were identified. First, defining appropriate levels of explanation is difficult. Different stakeholders, from technical teams to end-users, require different degrees of detail, and overly technical or overly simplified explanations can undermine understanding.

Second, selecting appropriate types of explanations is challenging, due to the subjective, context-dependent nature of user preferences, making personalized explanations so difficult to scale across diverse user groups.

Third, complex models such as deep neural networks pose explainability challenge. Techniques like figures, mathematical expressions, and SHAP values can be misleading to the end-users and are better suited for developers. Furthermore, interpreting some ML models are demanding even for data scientists, making it especially challenging to explain the decisions made by these models to non-technical users.

Fourth, explaining GenAI and LLMs presents a growing concern, as these models are considered mainly unexplainable. This raises risks around monitoring and human agency, adding to the insufficient organizational training and guidance on their use.

Fifth, a lack of continuous end-user testing was identified as a challenge. Since technical testing under MLOps practices well-established, but the continuous end-engagement and testing is often neglected.

Finally, risks associated with ignoring the data purpose and source were highlighted as an explainability challenging. Th experts shared that repurposing data without clear documentation can compromise both the accuracy of the AI outcomes and the ability to explain them responsibly.

### 4.3 Role of explainability in addressing the EU AI Act

The research regarding the third research question, how can explainability of AI systems address the transparency requirements of the EU AI Act, was performed in Paper IV. The research entailed in a lightweight approach to transform the transparency requirements of the AI Act to explanations of AI systems.

#### 4.3.1 An approach to address the transparency requirements of the AI Act

The AI Act mandates the high-risk AI systems be accompanied by instructions for use to satisfy transparency requirements (Article 13, EU AI Act). These instructions are expected to convey key details about the system, such as its intended purpose, capabilities, characteristics, and limitations (Article 13). The aim of the instructions for use is to enable deployers to use the AI system appropriately and make informed decisions regarding its use (Recital 72, EU AI Act). Table X summarizes instructions for use based on Article 13 and Recital 72 of the AI Act.

Table 6. Summary of instructions for use (EU AI Act)

|     |                                                                                                                        |
|-----|------------------------------------------------------------------------------------------------------------------------|
| Aim | Instructions for use should assist deployers in the use of the AI system and support informed decision making by them. |
|-----|------------------------------------------------------------------------------------------------------------------------|

|                                                         |                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Language                                                | Instructions for use should be made available in a language which can be easily understood by target deployers.                                                                                                                                                                                                                            |
| Format                                                  | Instructions for use should be accessible in an appropriate digital format or otherwise.                                                                                                                                                                                                                                                   |
| Quality                                                 | Instructions for use should include concise, complete, correct and clear information that is relevant and understandable to deployers.                                                                                                                                                                                                     |
| Aspects of AI systems explained in instructions for use | <ul style="list-style-type: none"> <li>• Intended purpose of AI system</li> <li>• Role of humans and AI system</li> <li>• Capabilities of AI system</li> <li>• Characteristics of AI system</li> <li>• Input data of AI system</li> <li>• Output of AI system</li> <li>• Limitations of AI system</li> <li>• Risks of AI system</li> </ul> |

Based on the transparency requirements outline above and the insights derived from the case study (Paper IV), we introduce a light-weight approach comprising a set of principles, practices, and guiding questions (Figure X). The key idea underlying this approach is to translate the AI Act's transparency requirements into concrete explanations of AI systems.

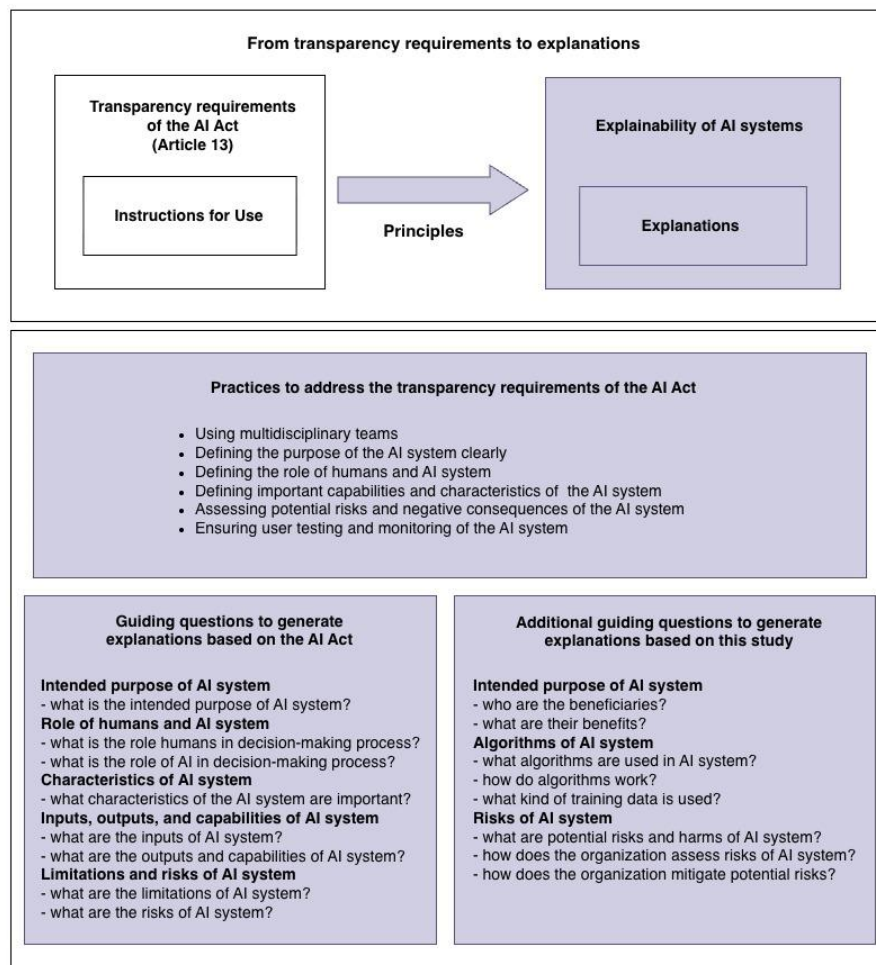


Figure 2. A lightweight approach to address transparency requirements of the AI Act

As part of the approach, we proposed three principles that can assist organizations for meeting the transparency requirements of the AI Act:

- **Principle 1:** The scope of explainability of AI systems is expanded from explaining operations and algorithms to cover all aspects of AI systems that are mandatory in transparency requirements and instructions for use of the AI Act.
- **Principle 2:** The context of the instructions for use is implemented as explanations that users can request from the AI system.
- **Principle 3:** Explanations must be relevant and understandable to users.

The approach suggests six practices (Fig. 2) to help organizations address the transparency requirements of the AI Act. The first five practices are primarily relevant to the requirements engineering process, while the sixth centers on user testing and monitoring, supporting the AI systems across their development and deployment stages. These six practices are explained in the following section when reporting the good practices that support the development of explainable AI systems.

### 4.3.2 Guiding questions and example of explanations

This part of the approach introduces guiding questions designed to help generate explanations that can be integrated into AI systems, aligned with the information required in instructions for use under the AI Act's transparency requirements. To illustrate their application, the guiding questions are demonstrated using a non-high-risk finance management application developed by our case organization for a Finnish bank. We used the bank's published transparency report as a source for example explanations. These guiding questions are intended to function as a practical checklist that organizations can use to address the transparency requirements of the AI Act. The example explanations for two aspects under the instructions for the use are added below and the example explanations for other aspects can be found in Paper IV.

*Intended purpose of the AI system* - The AI Act defines intended purpose as the use for which an AI system is designed by its provider, requiring providers to inform deployers of the purpose and proper usage. According to the case organization, it is important to clearly communicate the AI system's purpose, benefits, and beneficiaries to users and relevant stakeholders is essential to responsible AI development.

Table 7. Intended purpose of the AI app and example of explanations

| Guiding question | Explanation from the transparency report |
|------------------|------------------------------------------|
|------------------|------------------------------------------|

|                                                     |                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the intended purpose of the AI system?      | The intended purpose of the “finance management” app is to help users monitor their financial situation, adjust their consumption, and improve their saving habits.                                                                                                     |
| Who are the beneficiaries? What are their benefits? | <b>Users</b><br><i>In the short term</i> , the AI app makes daily money management easier and helps users with savings and spendings decisions.<br><i>In the long term</i> , it helps users to balance their finances, build up wealth and create a financial headroom. |
|                                                     | <b>Bank</b><br>The AI app is expected to have a positive impact on customer experience and customer satisfaction. These affect the duration of customer relationship and further the bank’s income from the customer relationship throughout the lifecycle.             |
|                                                     | <b>Society</b> – The bank’s hope is that AI app will have a significant impact on how people in Finland manage their daily finance.                                                                                                                                     |

*Role of humans and AI systems* – The AI Act requires instructions for use to specify human oversight measures, including the ability to disregard or override high-risk AI system outputs, and grants affected individuals the right to a clear explanation of the AI system’s role in decisions made. The industry experts stressed the importance of clearly distinguishing humans and AI roles in high-risk context. The experts noted that such explanations may be unnecessary for low-risk use cases, such as predicting parking availability or automatically blurring video backgrounds.

Table 8. Role of humans and AI app and examples of explanations

| Guiding question                                                         | Explanation from the transparency report [D6]                |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| What is the role of humans and AI system in the decision-making process? | Users can decide how to use the outputs of the AI app.       |
|                                                                          | The AI app does not recommend any actions for users.         |
| What is the role of users in the AI system?                              | Users can correct the transaction classification made by AI. |
| What is the role of AI in the AI system?                                 | AI classifies transactions as correctly as possible.         |

*Characteristics of the AI system* – Article 13 and Recital 72 of the AI Act require instructions for use to provide meaningful and understandable description of the characteristics of the AI system. The case organization highlights transparency, explainability, fairness, robustness, privacy, security, auditability, and trustworthiness as their foundational characteristics. The guiding question that can be used to generate explanations for the aspect characteristics of the AI system is: *what characteristics of the AI system are important to explain to users?*

*Inputs, outputs, and capabilities of the AI system* – Article 13 requires instructions for use to describe the AI system’s capabilities, including input data specification where relevant. These instructions should also help deployers interpret and appropriately use the system’s outputs. The experts emphasized that ethical AI development requires understanding not only the system’s capabilities, but also

user's domain expertise, their role and responsibilities in decision-making, and their ability to how to contest the outputs that they disagree with. The guiding questions that can be used to generate explanations for the aspect inputs, outputs, and capabilities of the AI system are: 1) *what are the inputs of the AI app?* 2) *what are the outputs and capabilities of the AI app?*

*Algorithms of the AI system* – The transparency requirements of the AI Act do not mandate explaining algorithms within instructions for use (Article 13). However, the transparency report for the finance management application still included explanations of its AI algorithmic components. Industry experts noted that the algorithms used in an system influence its explainability and emphasized the importance of understanding algorithmic precision and technical detail from a development perspective, and they clarified that such detailed need not be conveyed to end-users. The guiding questions that can be used to generate explanations for the aspect are: 1) *what algorithms are used in the AI system and how do they work?* 2) *what kind of training data is used?*

*Limitations and risks of the AI system* – Article 13 requires instructions for use to clearly explain the limitations of a high-risk AI system's performance, while Article 14 notes that understanding these limitations supports monitoring and helps identify anomalies or unexpected behavior. Industry experts widely emphasized the importance of understanding and communicating these limitations, viewing public disclosure as an ethical practice. The experts highlighted three key user-related risks: wrong outcomes produced by the AI system, overreliance on the AI system, and using outcomes of AI systems without critical assessment.

## **4.4 Practices to develop explainable AI systems**

The fourth research question of this thesis was to study what good RE practices can support defining explainability of AI systems. The answer for this question is based on all the papers (I-IV).

### **4.4.1 Practices to define explainability requirements**

In our Paper II, we studied what ethical guidelines organizations have defined to develop transparent and explainable AI systems. The analysis helped us proposing a model of explainability components with examples. The explainability components were created based on the explainability definition proposed by Chazette et al. (2021). The purpose of the model is to give a structured overview of explainability. The four components can be summarized with the following questions:



- Target groups – to whom to explain?
- Aspects – what needs to be explained?
- Context – when to provide explanations?
- Explainers – who explains?

Figure 3 shows the model of explainability components with examples for each component. Furthermore, we also presented the explainability components as a simple sentence i.e., a template (Figure 4). The purpose of the template is to assist practitioners to represent individual explainability requirements in a structured and consistent manner. The simple template is inspired by the user stories templates by Cohn (2006) which is widely used to represent functional requirements in agile development.

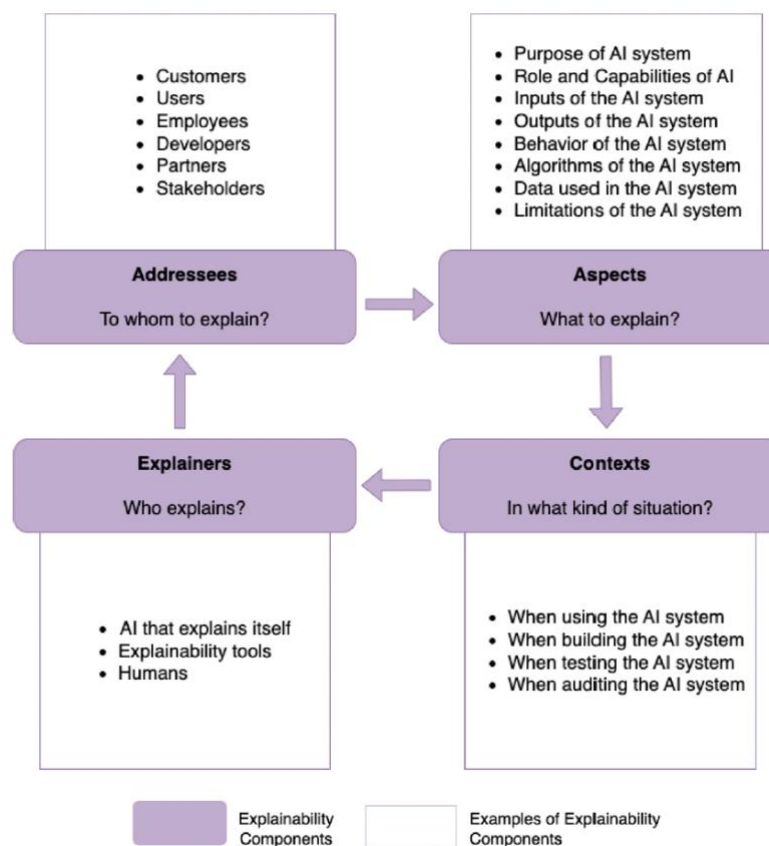


Figure 3. A model of explainability components

As a **<type of addressee>**, I want to get explanation(s) on an **<aspect>** of a **<system>** from an **<explainer>** in a **<context>**.

Figure 4. A template for representing individual explainability requirements

In the Phase 2 of our Paper II, we tested the model of explainability components (Fig. 3) with practitioners for a recruitment game AI system and used the data collected to define the explainability requirements for the system. We defined the following examples of explainability requirements by using the template (Fig. 4):

- As an applicant, I want to get explanation(s) from the recruiters on the *meaning and consequences of the results of the game* after using the game.
- As an applicant, I want to get explanation(s) from the recruiters on *how to interpret the scores of the recruitment game* after using the game.

We also shared our reflections on using the model of explainability components and template for defining explainability requirements in the form of lessons learned (detailed descriptions in phase of Paper II). We reformulated the lessons learned them into good practices to make it concrete for practitioners. Based on the lessons learned from our study, we suggest the following good practices to be used in the definition of explainability requirements of AI systems:

- Gain a good and shared understanding of the user and organizational processes that the AI system is supposed to support.
- Define clearly the purpose of the AI system and its role from the perspective of users and other stakeholders of the system.
- Analyze critically the potential risks and negative consequences of the AI system.
- Organize workshop(s) with a multidisciplinary team to capture different viewpoints on the explainability of the system.
- Use the model of explainability components to identify and analyze the explainability needs of stakeholders.
- Use a template to represent the explainability requirements of the AI system in a structured way.

#### **4.4.2 Good RE practices to develop explainable AI systems**

All four research papers of this thesis propose a set of practices that support the development of explainability of AI systems either based on literature review or empirical research. Based on the results of these studies, we consolidate a set of good practices that are useful in practice to develop explainable AI systems. In addition, some of these practices are also a part of the lightweight approach we proposed (Fig. 2) to support organizations in addressing the transparency requirements of the EU AI Act.

**Defining clearly the purpose of the AI system.** This is a common practice suggested by practitioners and highlighted in literature review of our studies. Case companies emphasized clearly defining the purpose of an AI system is foundational. Having a clear purpose not only helps ensuring customers' and users' trust but also

for building useful system for them. The recruitment game case illustrated that skipping this step can leave stakeholders to infer the benefits themselves, making subsequent explainability discussions fuzzy and difficult.

A clearly defined purpose serves as the foundation for the remaining practices and guides the overall development of the AI system. To further enhance transparency, organizations can identify the beneficiaries of the system along with the specific benefits they receive. Along with the benefits to users, the benefits for the company should also be explained with equal clarity and transparency, this helps the users to compare the value they receive against the value captured by the company. From the sustainability perspective, organizations should additionally consider explaining the broader benefits to society, nature, the climate, or people more generally.

**Identifying explainability needs.** This practice contains three interrelated practices: identifying stakeholders and their explainability needs, understanding business goals related to explainability from an organizational perspective, and establishing a shared understanding of users and organizational processes to provide context when identifying explainability needs. During the Phase 2 workshop (Paper II), our understanding about the recruitment game system was primarily based on our own interpretation, which limited the discussion of its explainability needs. When eliciting the explainability needs of the recruitment game system, we recognized that having a deeper, pre-established understanding of the system and its process would have enabled them to ask more richer follow-up questions and elicit explainability needs more effectively.

**Organizing workshops with multidisciplinary teams.** The case organizations emphasized the value of multidisciplinary teams, combining designers, social scientists, domain experts, and data scientists to incorporate diverse perspectives and develop understandable explanations. This was also reflected in the recruitment game workshop, where developers and recruiters contributed complementary system-level and organizational-level perspectives, enriching discussion on explainability. However, the absence of applicants, who were the actual end-users of the game, limited the range of perspectives considered, highlighting the value of involving end-users directly to validate existing and identifying additional explainability needs.

According to industry experts, multidisciplinary knowledge plays a crucial role in developing explainable AI systems. Bringing together designers, ML architects, and domain experts enables the integration of human, technical, and application-specific knowledge throughout the development process. The experts further suggested involving ethicists, political scientists, social scientists, and anthropologists to help assess the broader consequences of the system.

Furthermore, from the AI Act perspective, multidisciplinary teams are particularly beneficial in three areas: defining the intended purpose of the AI system, clarifying the respective roles of users and the AI system, and assessing potential risks and harms associated with the system.

**Using the model and template when defining the explainability requirements.** During the recruitment game workshop, the model of explainability components structured the identification of target groups, aspects, context, and explainers of the AI system. Using the explainability components during the workshop with multidisciplinary teams is a practice that supports the identifying explainability needs. The template provides a structured way to represent the explainability needs in the form of explainability requirements.

**Defining the role of humans and AI system.** This practice contains two key tasks. First, it is important to determine the role of humans in the decision-making process. According to the AI Act, humans must be able to truly monitor the operations of high-risk AI system, including identifying and addressing anomalies, dysfunctions, and unexpected performance. Additionally, in the case of high-risk AI systems, humans must retain the ability to disregard, override, or opt not to use the AI system's output altogether. Second, it is important to define the role of the AI system in the decision-making process, which can generally take the form of either supporting human decisions or automating tasks otherwise performed by humans. The experts further emphasized that, regardless of the AI system's role, responsibility for the consequences of decisions ultimately rests with the people and organizations deploying the AI system.

**Defining important capabilities and characteristics of the system.** This practice lies at the heart of the typical requirements definition process. Capabilities of an AI system corresponds to the functional requirements or use cases within requirements engineering and software engineering. User stories offer one light-weighted technique for capturing key functional requirements concisely and clearly from a user-centered perspective. Characteristics of an AI system, by contrast, refers to quality requirements within requirements engineering and software engineering. The AI Act identified three such requirements for high-risk AI system, namely accuracy, robustness, and cybersecurity, while the consulting companies involved in our study additionally identified transparency, explainability, fairness, robustness, privacy and security as foundational to trustworthy and sustainable AI.

**Analyzing critically the potential risks and negative consequences of the AI system.** Risk assessment not only supports innovation but that compliance with the AI Act can also act as a competitive advantage. Assessing the potential risks

and negative consequences of AI systems involved several key activities: 1) conducting risk assessment workshops with multidisciplinary teams, 2) engaging a senior facilitator who understands the context, is willing to challenge the team on AI-related risks, and is prepared to probe the issue in depth, 3) involving senior experts with strength, expertise, experience, and willingness to conduct risk assessment and take accountability, 4) prioritizing the analysis of typical AI risks, such as incorrect outcomes and overreliance on AI systems

**Ensuring user testing and monitoring of the AI system.** User testing benefits organizations aiming to build transparent and explainable AI systems and should be given equal importance alongside technical testing. Methods such as prototyping, interviews, think aloud sessions, and focus groups are among the practices useful for testing AI system explainability.

The experts noted that monitoring is an ongoing process rather than a one-time activity, since AI systems cannot remain static and require periodic training with new data. The expert shared the five key tasks involved in monitoring: tracking system performance over time, assessing the quality and source of input data, evaluating output results, monitoring how results are presented through user interfaces, and overseeing how the underlying ML models are trained or retrained.

## 5 Discussion

In this section, the results are discussed and the answers to the research questions are presented. The main findings of the study are compared with the results of previous research. After that, the limitations and threats to validity are summarized, followed by the review of the implications for researchers and practitioners.

### 5.1 Answers to research questions

In this subsection, the results of this thesis are discussed and answers to the four research questions are presented. The discussion follows the order of the research question stated in Section 3.1

#### 5.1.1 Quality requirements of AI systems

The first research question of this thesis was to identify what quality requirements organizations emphasize in AI ethical guidelines.

Firstly, the analysis of the case companies' AI ethical guidelines revealed a strong focus on addressing potential ethical issues, particularly accountability, explainability, fairness, privacy, and transparency. These findings closely align with ethical issues identified in existing literature, including anonymity (Doyle et al. 2014), fairness (Jones et al. 2015, Palm, 2009, Royakkers et al. 2018), security (Jones et al. 2015, Palm, 2009, Pieters, 2011, Royakkers et al. 2018), privacy (Etzioni & Etzioni, 2016), transparency (Elia, 2009, Jones et al. 2015, Royakkers et al. 2018), and trust (Elia, 2009, Jones et al. 2015, Pieters, 2011). This suggests that the case companies' concerns reflect broader issues already known in the field and still tailoring the issues that are relevant for the AI systems they develop in their organizations.

The case companies' guidelines also closely matched with guidelines from three established expert groups (European Commission, 2020, IEEE Ethically Aligned Design, 2019, SIIA, 2017). For example, the companies' accountability guidelines aligned with the expert groups' well-being and responsibility principles. Both the expert groups (European Commission, 2020, IEEE Ethically Aligned Design, 2019) and our case study's interviewees also stressed the need to clearly define the

purpose of the AI system. This overlap suggests that organizational guidelines are shaped by, or at least aligns with, established expert guidelines.

All case companies had defined guidelines on transparency. These guidelines focused on being open about AI use, clearly communicating data use practice, and prioritizing transparency to build user trust. This supports earlier work by Cysneiros et al. (2018, 2020) and Horkoff (2019), who also identified transparency as a key quality requirement in AI and ML systems. Thirteen organizations treated explainability as a core part of transparency. This matches findings by Chazette et al. (2019) and Chazette and Schneider (2020), who found that explanations improve transparency. However, this thesis adds a new insight that, distinguishing clearly between transparency and explainability was difficult, not just as a practice (Chazette et al. 2019), but also in how organizations defined described these terms in practice. This points to a need of clearer definition in future guidelines.

Finally, the main reason organizations gave for using transparency and explainability was to build and maintain trust. This matches earlier studies linking explainability to trustworthy AI (Chazette et al. 2021, Drobotowicz et al. 2021) and Zieni and Heckel's (2021) finding that transparency helps build user trust. It also aligns with Cysneiros et al. (2018) and Habibullah and Horkoff (2021), who found that trust is a critical quality requirement in autonomous and machine learning systems. Together, these findings suggest that trust connects transparency and explainability across both research and practice, making it a central concern in ethical AI development.

### **5.1.2 Industry experts' perception of explainability**

The second research question of this thesis was to explore the perceptions of the industry experts on explainability of AI systems. The results covered why explainability is seen important by industry experts and what are the challenges involved in implementing explainability of AI systems.

The industry experts of the case study said that explainability mainly helps users understand AI systems. They also noted that other stakeholders, such as lawyers, ML engineers, designers and developers, often need to understand AI systems too, even though for different reasons. These stakeholders want to understand several aspects of AI systems, including results, algorithms, decision-making processes, risks and limitations. This case study result support the model of explainability components proposed in Paper II. They provided real-world evidence for the model's two questions: who needs explanations, and what should be understood.

The findings also show that AI Act compliance is becoming a major driver of explainability, especially for high-risk systems. Explaining AI decisions and ensuring human accountability are essential for building trustworthy systems. This matches Wagner et al. (2025), who found that AI Act compliance improves

trustworthiness but requires significant effort to implement. It also matches Scantamburlo et al. (2024), who founds that organizations face technical, organizational, and cultural challenges when complying with the Act. Overall, these results suggest that explainability is not only an ethical practice. It is also becoming a practical tool that helps organizations comply with the AI Act.

From a development perspective, industry experts said that choosing the right type and level of explanations is a common challenge. This matches the work by Dhanorkar et al. (2021) and Liao et al. (2020), who highlighted the difficulty in deciding how explanations should be presented and how much detail different stakeholders need. This shared challenge across studies suggests that explanation design remains an unresolved problem, not just an issue specific to one organization.

Finally, a lack of interest and resource was identified as a major barrier to developing explainable AI systems. Experts said that this often comes from limited organizational focus on ethical AI development. This finding is supported by a prior work by Habiba et al. (2025) which also reported limited computational resources for generating accurate explanations, and Liao et al.'s (2020) UX designers said product teams often lack the time and resources to build explainability into AI systems. All in all, these findings suggest that limited resources and low prioritization are common industry-wide barriers to explainability, not problem unique to the case organization in this thesis.

### **5.1.3 Explainability in addressing transparency requirements of the AI Act**

The third goal of this thesis was to investigate how explainability support in addressing the transparency requirements of the AI Act. The study used instructions for use in the transparency requirement (Article 13) as the foundation for proposing a lightweight approach.

The findings show that giving users clear and informative explanations can help organizations meet the transparency requirements of the AI Act. The lightweight approach proposed in this thesis can be especially useful for SMEs and startups, which often have fewer resources than larger organizations. During the workshop (Paper IV), experts compared the aspects listed in the AI Act's instructions for use (European Commission, 2024) with their own experience and they agreed that these aspects are important for both explaining and understanding AI systems. So, the lightweight approach supports the practical value of the AI Act's requirements.

The results also show that clearly defining the roles of humans and AI systems is an important practice for AI Act compliance. This supports the model of explainability components (Paper II), which identifies the role of the AI system as something that needs to be explained to stakeholders. This practice of the proposed lightweight approach connects to Article 14 of the AI Act (European Commission,



2024), which requires effective human oversight. Therefore, this suggests that clearly defining humans and AI system roles is not just a good practice, it is also necessary to meet legal requirements.

Furthermore, the findings also point to having multidisciplinary teams as an important practice for creating explanations that support AI Act compliance. This is especially helpful for defining human and AI roles and for conducting risk assessments. This practice of having multidisciplinary teams and its usefulness when developing AI system is a recurring result in all four publications of this thesis. This suggests the practice works well across different contexts, not for just one context.

Finally, a key finding suggests that identifying risks and negative consequences is an important practice. Industry experts pointed to three key risks that should be recognized during the AI system development, they are, AI systems producing wrong or incorrect outcomes, overreliance on the AI system, using outcomes of AI systems without critical assessment. This practice matches Article 9 of the AI Act (European Commission, 2024), which requires continuous risk management, including risk identification, analysis, and evaluation, throughout the lifecycle of a high-risk AI system. However, other researchers raise concerns here, Cheatham et al. (2019) and Novelli et al. (2024) argue that traditional risk management methods are not enough for AI systems, because AI risks are complex and constantly changing. Building on this, the results of this thesis suggest that explaining risks to stakeholders can be a useful first step in risk management. This can help organizations spot both common and emerging risks that traditional methods might miss.

#### **5.1.4 RE practices for defining explainability of AI systems**

The fourth research goal of this thesis suggests a set of good practices that support in defining explainability of AI systems. We discussed three good practices in the previous subsection when addressing how explainability supports fulfilling transparency requirements of the AI Act.

A core motivation was to provide practitioners with actionable solutions rather than purely theoretical guidance. Drawing on longitudinal action research on RE process improvement (Kauppinen et al. 2004), this thesis recommends that practitioners begin by adapting a set of existing RE and user-centered design practices, rather than developing entirely new methods, since tailoring familiar practices tends to lower the barriers to adopt explainability in AI projects.

Among these practices, clearly defining the purpose of the AI system emerged as foundational and the case organizations (Paper I) emphasized that defining the purpose of AI system as a first step to ethical AI development. A clearly articulated purpose was found to guide and simplify subsequent work, particularly for

multidisciplinary teams aiming to define explainability requirements. This reinforces the logic proposed in this thesis, purpose must be defined well before identifying the explainability needs.

Next, having multidisciplinary teams when developing AI system is a recurring practice in our studies. The workshop study (Paper II) supports our first case study findings (Paper I) that organizations value multidisciplinary teams for bringing in different viewpoints. This also aligns with broader RE research. Nuseibeh and Easterbrook (2000) recommended interviews as core technique for gathering requirements. Maalej et al. (2023) also recommend interviews and workshop for responsible AI development. Together, these findings suggest that having multidisciplinary teams is not just beneficial but foundational across both explainability specific research and general responsible AI literature.

Then, systematically identifying explainability needs, was directly supported by the model of explainability components (Paper II), which structures explainability need identification around four elements: target groups (addressees), aspects to be explained, the context in which the explanations are needed, and explainers who provide them (Chazette et al. 2021). The workshop study demonstrated this model in a step-by-step process, first identifying stakeholders, then relevant aspects, and finally context and explainers. Moreover, this process surfaced an important improvement, participants identified humans, not just AI systems, as explainers. This finding is consistent with Köhl et al. (2019), who similarly note that human experts can provide explanations. This finding necessitated updating the original model to include humans as explainers, illustrating how empirical validation can help improve frameworks rather than simply confirming them.

Once explainability needs are identified, structured templates were found to support consistent representation of individual requirements. This finding aligns with Sommerville and Sawyer's (2004) view that standard templates make requirements easier to write, read, and communicate. This thesis extends the view specifically to explainability, proposing a simple template resembling user stories. Furthermore, this finding is consistent with Inayat et al.'s (2015) recommendations for agile RE, and Maalej et al.'s (2023) guidance for responsible AI suggesting that lightweight, familiar formats are particularly well suited to explainability's cross-disciplinary and often non-technical stakeholders and end-users.

Finally, analyzing potential risks and negative consequences as an important practice. While this thesis did not observe a direct impact of risk analysis on explainability requirements definition, findings indicated that negative consequences could shape core functional requirements first, which subsequently could influence explainability requirements. This practice closely relates to misuse cases, an established RE technique for identifying hostile or negative use scenarios (Alexander, 2003). A case organization's ethical guidelines also supported this, calling for organizations to think about the system's impact on society, not just on

users. In the recruitment game case, participants raised concerns about black-box behavior and bias. These concerns are supported by Dattner et al. (2022), who highlight the legal and ethical risks of using AI in hiring. This suggests that risk analysis while not directly defining explainability requirements, plays an important role in discovering the risks that explainability requirements must eventually address.

All in all, these practices, starting with existing methods, defining purpose of AI system, engaging multidisciplinary teams, systematically identifying explainability needs, using structured templates, and analyzing risks form a cohesive approach to defining explainability requirements and incorporating explainability to AI systems. These practices are grounded in both the empirical findings of this thesis and consistent with established RE and responsible AI literature.

## **5.2 Limitations and threats to validity**

This section discusses the limitations and threats to validity of this thesis. They are organized around common validity concerns as defined in case study and qualitative research methodology (Runeson & Höst, 2009; Yin, 2018). In addition, the scope-related limitation of the thesis is summarized.

### **5.2.1 Generalizability and External Validity**

A recurring limitation across the empirical studies (Papers I, II, and IV) is the relatively small and context-specific nature of the case organizations. Paper I involved a low number of case companies, though these represented three distinct application domains and were identified by an AI expert as forerunners in ethical AI development. This suggests that their practices offer transferable insights despite the limited sample size. This aligns with established guidance that case study generalizability depends less on sample size and more on the richness and relevance of the cases selected (Flyvbjerg, 2006). Paper II's analysis of 16 organization's ethical guidelines was similarly region specific because majority of the organizations being Finnish or Finland-based. However, comparison with the three global organizations in the sample revealed no significant differences, suggesting findings may extend beyond the Finnish context. Paper IV relied on a single case organization, limiting the direct applicability of findings to other organizations, though the organization's significant cross-domain AI development experience allowed informants to reflect on challenges and practices that could be relevant to less mature organizations. Across these studies, external validity was further supported through data triangulation (Patton, 1999), including document analysis (Papers I, II, IV), workshops (Papers II, IV), and extended interview durations (80-

90 minutes in Paper IV), which helped enrich the understanding beyond what interviews alone could provide.

### **5.2.2 Reliability and Researcher Bias**

Researcher bias represents a consistent threat of this thesis work. To mitigate this, each study employed researcher triangulation (Patton, 1999). Papers I and II involved multiple researchers interview design, data collection, independent coding, and multiple peer meetings iteratively to resolve the disagreements in the codes. Similarly, Paper III used triangulation between authors with different backgrounds (requirements engineering and technical perspectives) to counter bias in the literature selection and analysis process. Paper IC addressed the risk of shared bias between long-term collaborating researchers by involving a third researcher to critically review the chain of evidence from data to conclusions. Despite these measures, the potential for interpretive bias, particularly in single researcher led processes such as the literature selection in Paper III, remains an limitation.

### **5.2.3 Construct Validity**

Construct validity concerns primarily relate to sample size and the interpretation of key concepts (Runeson & Höst, 2009). Paper I interviewed only on representative per case company. This is addressed through triangulation with ethical guideline documents, though this restricted the study to a company-level perspective rather than capturing how guidelines are applied within individual AI projects. The analysis of Paper III was based on twelve publications, focused specifically on explainability within the RE field, meaning relevant work from other fields such as XAI may not have been captured. This is an inherent trade-off in literature reviews with narrowly scoped search strategies (Kitchenham & Charters, 2007). Paper IV faced the additional challenge that key terms in this research area, such as transparency and explainability, and user, carry multiple meanings, creating a risk of inconsistent interpretation among interviewees. This was mitigated by tailoring interview questions to each participant's area of expertise.

### **5.2.4 Internal Validity**

Internal validity was a particular concern in Paper IV, where the proposed relationship between explainability aspects and the AI Act's transparency requirements may not be fully correct. To strengthen this proposal, the first two authors iteratively compared the AI Act's requirements against the raw data, while third experienced researcher reviewed the proposed approach throughout its

development. Moreover, the research team's background is mostly computer science and software engineering rather than law, a legal expert was additionally consulted to verify the interpretation of AI Act requirements, a practice that is consistent with recommendations for triangulating domain-specific expertise in interdisciplinary research (Patton, 1999).

#### **5.2.5 Scope-related limitations**

Several study-specific scoping limitations occurred in this thesis. In Paper II's workshop phase, the evaluation was limited to a single workshop without subsequent validation with recruiters and developers. This means practical usefulness of the proposed template remains uninformed which reflects the recommendation that empirical evaluations of candidate solutions should proceed incrementally, from expert opinion to controlled case studies to industry practice (Wohlin & Runeson, 2021). Similarly, applicants which is one of the primary user groups of the recruitment game were excluded to encourage open discussion among recruiters, a decision that facilitated candid discussion, but left applicant perspectives underrepresented. The recruitment game itself originated as a student project and did not represent a typical organizational project, but selection was considered appropriate given the sensitivity and complexity of AI use in hiring contexts.

In Paper III, the literature search was limited to major RE publications forums, meaning the review does not comprehensively capture explainability-related work published outside RE field. This work could serve as a starting point for future broader systematic reviews by having the included publication set as a starting point for snowballing (Wohlin, 2014). Finally, Paper IV has purposeful selection of informants which is selected specifically for their involvement in AI-related organizational activities. This means the findings primarily reflect the perspectives of industry experts rather than end-users. Additionally, the lightweight approach proposed in Paper IV has not yet been evaluated in a real-world project, which is identified as a direction for future work.

Overall, these limitations suggest that all individual studies in this thesis are constrained by sample size, single organization focus and inherent researcher subjectivity in qualitative analysis. The consistent use of triangulation both in data sources and among researchers, across all four studies strengthens the overall reliability and credibility of the thesis's combined findings.

### **5.3 Implications for research**

This thesis contributes to RE research by demonstrating that explainability is closely interconnected with other quality requirements, such as transparency,

trustworthiness. Explainability should therefore not be treated as an isolated concept. Future RE research should examine the interdependencies more closely, particularly in high-risk AI domains. In such domains, trade-offs between and explainability and other qualities, such as performance, privacy, or system complexity, can carry significant consequences.

This thesis also found that although many models and frameworks for defining explainability requirements exist, most remain in research field without transferring the knowledge to practice. This points to a gap in RE research where theoretical contributions need stronger empirical evaluation. The model and template proposed in this thesis address part of this gap, but future work should continue testing tools with practitioners in real project settings.

This thesis also opens two further research directions. First, by connecting explainability to the EU AI Act, it illustrates emerging research area, translating legal requirements into concrete system requirements. As AI regulation evolves, RE research should develop and test similar approaches across different regulatory contexts. Second, this thesis identified a gap between how industry experts and customers perceive explainability, an area that requires further research. Future studies should investigate how explainability needs can be better elicited from non-technical stakeholders, and how awareness of its value can be integrated into requirements elicitation practices.

## **5.4 Implications for practice**

The findings of this thesis carry several practical implications. Organizations should treat explainability as a core quality requirement, rather than an optional add-on, given its role in supporting trust, transparency, and regulatory compliance. Neglecting this could lead to ethical and legal risks, particularly under the EU AI Act. The model of explainability components and the template for representing explainability requirements proposed in this thesis offer practitioners concrete tools for identifying who needs explanations, what should be explained, in what context, and who should provide them. Applying these tools early, during requirements engineering, can help teams avoid costly redesigns later in development. This thesis also underscores the value of multidisciplinary teams involving designers, domain experts, and ethicists alongside developers helps ensure explainability requirements reflect diverse perspectives.

This thesis further offers guidance for regulatory compliance and stakeholder engagement. The lightweight approach proposed gives organizations, particularly SMEs and startups with limited resources a practical way to address the transparency requirements of the EU AI Act. Finally, since customers were found to underestimate the importance of explainability compared to practitioners, organizations should invest in raising awareness among customers and business

stakeholders. Communicating the value of explainability can strengthen user trust and helps organizations with compliance with emerging AI regulations.

## 6 Conclusions

This chapter first summarizes the main contributions of this thesis, and then, future research directions are discussed.

### 6.1 Contributions of the research

This thesis explored how requirements engineering can support the definition of explainability of AI systems. The findings show that organizations consistently prioritize accountability, explainability, fairness, privacy, and transparency as critical quality requirements in their AI ethical guidelines, with explainability emerging as tightly coupled to both transparency and trustworthiness. In addition, clearly defining the purpose of the AI system and analyzing its impacts were identified as important enablers for multidisciplinary teams addressing the ethical issues during AI development.

From the industry perspective, explainability was consistently recognized by practitioners as essential for improving understandability of AI systems and building user trust. However, a notable gap emerged between practitioners and customers, as the latter often lacked awareness on importance and role of explainability. This highlights the need for more awareness among customers alongside technical solutions.

The thesis further demonstrated that explainability can serve as a practical mechanism for achieving compliance with the transparency requirements of the EU AI Act. A lightweight approach, comprising principles, practices, and guiding questions, was proposed to translate these regulatory requirements into concrete explanations covering an AI system's purpose, risks, and limitations which are applicable to both high-risk and non-high-risk AI systems.

Finally, this thesis addressed a critical gap identified in the literature. While various approaches, models, and frameworks for defining explainability requirements exists, most of them are not evaluated through empirical studies. To address the research gap, this thesis contributes a model of explainability components, structured around identifying target groups, aspects, contexts, and explainers, along with a structured template to represent explainability requirements. In addition, the thesis presents a set of good practices centered on defining purpose of AI system, analyzing potential negative consequences, and



engaging multidisciplinary teams. In total/Collectively, these contributions show that existing RE and user-centered design practices can be significantly tailored to support the definition of explainability requirements in practice.

Overall, this thesis contributes to both the theoretical understanding and practical implementation of explainability of AI systems within requirements engineering. Explainability is shown to be more than an ethical and regulatory obligation. It is also a quality requirement that can be systematically defined and practically achieved.

## **6.2 Future Work**

The findings of this thesis open several research directions for future research. One important direction is to conduct case studies that examine how ethical guidelines, transparency requirements, and explainability requirements are applied in real AI projects. These studies would compare how development teams use these guidelines and requirements in practice, the challenges they face, and the positive experiences they report.

Another key direction is to evaluate the solutions proposed in this thesis with practitioners. This includes assessing the usability and usefulness of the model of explainability components and the explainability template in real project contexts. It also includes evaluating the lightweight approach proposed for addressing the AI Act's transparency requirements, particularly by involving users in assessing explanations generated through the approach. Such evaluations would help identify both the benefits and practical challenges of applying these solutions in industry.

A related long-term goal is to study how explainability requirements can support the testing of AI systems. Future work should investigate which practices are most useful at different stages of AI system development, from early requirements definition to testing and validation.

Future research should also examine the broader impact of the AI Act on AI system development. This includes a closer focus on risk management and monitorability requirements, both of which are central to the AI Act but were not fully explored in this thesis.

Finally, future work should connect the findings of this thesis more closely with research in the field of explainable AI (XAI). This includes analyzing how explainability requirements are approached in XAI research and identifying how this knowledge can be used to strengthen explainability practices in requirements engineering field.

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